European Commission

Realising the potential of the Outermost Regions for sustainable blue growth

ANNEX 12 TO THE FINAL REPORT

THE BLUE ECONOMY IN THE MACARONESIA SEA BASIN

Written by COGEA srl,
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Contents

Mapping economic activities **2**

1 Identification and mapping of economic activities 2
2 Synergies and tensions between maritime activities 11
3 Best practices and flagship projects 15
4 Gap analysis 20

Mapping actors and maritime strategies **24**

5 Mapping public authorities 24
6 Summary of existing maritime strategies 27
7 Mapping main stakeholders 30

Mapping Blue Growth and assessing the potential **33**

8 Growth drivers and barriers 33
9 Future potential for mature and pre-developing activities 38
10 Investments opportunities 48
11 Most relevant maritime activities at sea basin level 48

Identifying public and private intervention **51**

12 Potential financial, legislative and non-legislative measures for stimulating growth 51
13 Position of the EU ORs in the Macaronesia sea basin 57
14 Conclusions on concrete measures by type and timeframe 60

List of interviewees **69**

List of acronyms

CAGR	Compound annual growth rate
CAP	Common Agricultural Policy
CFP	Common Fishery Policy
EEZ	Exclusive Economic Zone
EIB	European Investment Bank
EDF	European Development Fund
EMFF	European Maritime and Fisheries Fund
ERDF	European Regional Development Fund
FDI	Foreign Direct Investment
FP	Formación Profesional
GDP	Gross Domestic Product
GVA	Gross Value added
SFPA	Sustainable Fisheries Partnership Agreement
ICCAT	International Commission for the Conservation of Atlantic Tunas
IEO	Instituto Español de Oceanografía – Spanish Institute of Oceanography
ITC	Instituto Tecnológico de Canarias – Canarias Technological Institute
IUU	Illegal, Unreported and Unregulated fishing
LNG	Liquefied Natural Gas
MAC	Transnational cooperation program Madeira, Azores and Canarias
MCS	Monitoring, Control and Surveillance
MMCM	Marine and Maritime Cluster of the Macaronesia
MSP	Marine Spatial Planning
OR	Outermost Region
POSEICAN	Former programme of options specific to the remote and insular nature of the outermost regions for the Canary Islands.
PLOCAN	Plataforma Oceanográfica de Canarias – Canarias Oceanographic Platform
RFMOs	Regional Fisheries Management Organisations
SWOT	Strengths, Weaknesses, Opportunities and Threats
SWW AC	South Western Waters Advisory Council
ULL	Universidad de La Laguna – University of La Laguna
ULPGC	Universidad de Las Palmas de Gran Canarias – University of Las Palmas de Gran Canarias

Introduction

Macaronesia is the name of a group of Atlantic archipelagos that extend outwards from the South-West of Europe to the North-West of Africa. The region comprises four main archipelagos: Azores, Madeira, Canary Islands and Cape Verde. Historically the Macaronesia region is defined in terms of geographical location and geological peculiarities. Although the current study is limited to the EU's OR the scope of the influence of these ORs goes beyond their political limits, straddling political, economic and research/development and innovation matters with Cape Verde. Consequently, the present summary includes the republic of Cape Verde.

There is an increasing awareness of the importance of the Macaronesia archipelago and its strategic position in the Atlantic. The sea basin is of strategic interest due to its political, geographical and cultural links to EU, Africa and South America. The economy of the region is highly dependent on the services sector, particularly on tourism, while primary sectors although having a limited presence in the GDPs have a strategic importance in food security and regional culture. The sea basin is particularly relevant for maritime transport, ports services, coastal and maritime tourism. It provides natural advantages for aquaculture, blue biotechnology and blue renewable energies. Advanced R&D in the ORs is an asset for the development of the diverse sectors. Traditional activities such as fishing have decayed in the last years in fleet size and processing capacity but play a key role in Cape Verdean economy. The EU has in place a transnational cooperation program (ERDF's MAC program) that includes Cape Verde. Substantial investments in the ORs have been carried out with support of the ERDF and, in the case of fisheries and aquaculture, through the structural fund for fisheries (currently EMFF). It is worth mentioning that the ORs' private sectors are carrying out efforts to promote Blue Growth through the Maritime Cluster of the Macaronesia, where Cape Verde is also taking part.

The present summary focuses on those marine and maritime sectors with presence in two or more archipelagos.

Mapping economic activities

1 Identification and mapping of economic activities

Table 1 — GVA of marine and maritime activities

Group	Sector	Maturity vs. Pre-maturity	GVA (1000 €)					CAGR
			2010	2011	2012	2013	2014	
Living resources	Fisheries	M	65 342	49 526	51 647	52 261	61 317	-7.18
	A03.11, A03.21, C10.20 Fishing and aquaculture	M	53 039	40 483	42 288	40 256	61 317	-8.78
	Fishing and aquaculture in Cape Verde	M	12 303	9 043	9 359	12 006	n.a	-0.81
Non-living resources	Non-living resources	M	63 991	61 835	55 773	54 986	56 308	-3.15
	B08.12 Extraction of aggregates	M	4 656	4 221	1 019	1 282	1 221	-28.44
	Desalination	M	59 335	57 614	54 754	53 704	55 087	-1.84
Shipping	Shipping	M	683 273	680 778	662 950	656 985	672 582	-0.39
	H50.10 Maritime transport	M	267 041	264 122	257 665	254 066	262 883	-0.39
	H50.20 Ports (including dredging)	M	416 232	416 656	405 285	402 920	409 699	-0.39
Shipbuilding	H30.11, H30.15 Ship building and repair	M	48 921	76 629	68 272	77 017	n.a	16.33
Tourism	Tourism	M	2 857 917	2 722 213	2 945 356	3 088 950	2 956 925	2.63
	I55, I56, H51 Coastal tourism	M	2 621 439	2 483 260	2 687 565	2 816 977	2 953 030	2.43
	H50.10 Sea and coastal passenger water transport (cruise share)	M	4 026	3 351	3 262	3 242	3 895	-5.27
	Tourism in Cape Verde	M	232 452	235 602	254 529	268 731	n.a	4.95
Marine renewable energies	Wave energy	M	3	1	1	0	2	-8.77

Table 2 - Employment in marine and maritime activities

Group	Sector	Maturity vs. Pre-maturity	Employment					CAGR
			2010	2011	2012	2013	2014	
Living resources	Fisheries	M	5 856	5 039	5 136	5 229	5 856	0.00
	A03.11, A03.21, C10.20 Fishing and aquaculture	M	5 856	5 039	5 136	5 229	5 856	0.00
	Fishing and aquaculture in Cape Verde	M	n.a	n.a	n.a	n.a	n.a	
Non-living resources	Non-living resources	M	1 138	1 116	1 084	1 031	1 049	-2.01
	B08.12 Extraction of aggregates	M	130	122	100	62	57	-18.63
	Desalination	M	1 008	994	984	969	992	-0.40
Shipping	Shipping	M	10 152	9 361	8 850	9 681	9 958	-0.48
	H50.10 Maritime transport	M	2 788	2 589	2 614	2 759	3 000	1.85
	H50.20 Ports (including dredging)	M	7 365	6 772	6 236	6 922	6 959	-1.41
Shipbuilding	H30.11, H30.15 Ship building and repair	M	1 037	1 084	1 184	1 150	44	3.51
Tourism	Tourism	M	113 959	111 275	116 711	115 002	119 549	1.20
	I55, I56, H51 Coastal tourism	M	113 856	111 241	116 675	114 964	119 409	1.20
	H50.10 Sea and coastal passenger water transport (cruise share)	M	103	34	35	37	140	8.02
	Tourism in Cape Verde	M	n.a	n.a	n.a	n.a	n.a	
Marine renewable energies	Wave energy	M	0	0	0	0	0	

Table 3 - Turnover of marine and maritime activities

Group	Sector	Maturity vs. Pre-maturity	Turnover (1000 €)					CAGR
			2010	2011	2012	2013	2014	
Living resources	Fisheries	M	204 402	151 944	156 405	147 499	240 620	4.16
	A03.11, A03.21, C10.20 Fishing and aquaculture	M	204 402	151 944	156 405	147 499	240 620	4.16
	Fishing and aquaculture in Cape Verde	M	n.a	n.a	n.a	n.a	n.a	
Non-living resources	Non-living resources	M	179 821	164 380	156 047	157 829	134 528	-7.00
	B08.12 Extraction of aggregates	M	15 302	13 298	5 837	3 490	3 520	-30.74
	Desalination	M	164 520	151 082	150 210	154 339	131 007	-5.54
Shipping	Shipping	M	1 420 936	1 402 424	1 419 349	1 386 951	1 341 305	-1.43
	H50.10 Maritime transport	M	546 817	549 869	543 656	469 224	496 480	-2.39
	H50.20 Ports (including dredging)	M	874 120	852 555	875 693	917 727	844 826	-0.85
Shipbuilding	H30.11, H30.15 Ship building and repair	M	102 306	160 730	133 246	166 550	2 108	17.64
Tourism	Tourism	M	6 327 092	6 314 243	6 578 643	6 869 134	955 182	2.08
	I55, I56, H51 Coastal tourism	M	6 179 135	6 167 010	6 433 837	6 747 411	824 834	2.22
	H50.10 Sea and coastal passenger water transport (cruise share)	M	147 957	147 233	144 806	121 723	130 349	-3.12
	Tourism in Cape Verde	M	n.a	n.a	n.a	n.a	n.a	
Marine renewable energies	Wave energy	M	7	2	3	0	5	-5.02

Notes:

Data in this table is provided in thousand Euros. Data for the EU's OR comes from the OR reports. In the case of Cape Verde, GVA data for fisheries and aquaculture and services sector are published by the Banco de Cabo Verde, and are only available until 2013. This bank provides the data in Escudos, so conversion into Euros has been carried out. Data on employment and turnover for fisheries, aquaculture and services are not available. GVA of the Cape Verdean touristic sector has been estimated by aggregating the data provided by the Banco de Cabo Verde on accommodation and restaurants with the data on transport sectors. The fisheries (including processing) and aquaculture analysis has been conducted for the period 2010-2013 due to the lack of data for Cape Verde in 2014. In the case of ship building and repair data for Canary Islands is not available for 2014, given the high relevance of this archipelago in these activities the analysis has been conducted for the period 2010-2013. Regarding tourism, the period of analysis is 2010-2013 due to the lack of tourism data for Cape Verde in 2014.

1.1 Fisheries and aquaculture

1.1.1 Overview

Fisheries

Marine commercial fishing has a long tradition in the Macaronesia sea basin. The activity is characterised by the predominance of small scale activities, with vessels smaller than 12 m in length. The fishing activity has shown a decreasing trend in number of vessels and fishermen due to the reduction of the ORs' fleets aimed at adjusting fleet capacities and fishing possibilities. The Portuguese ORs rely on deep sea species such as black scabbard fish, while for the Canary Islands the predominant catches are those of small pelagics. Tuna resources are also of high importance for the ORs because of their high economic value. Quotas of bluefin are allocated to the regions in the context of the TAC established annually by ICCAT. This is an important driver for the sustainability of the fishing activity. In turn, the Canary Islands fleets are highly dependent on access to the waters of the Western and Northern Africa countries and thus the Canary Island sector is highly sensitive to political decisions and negotiations on sustainable fishing partnership agreements (SFPA). The fishing activity in Cape Verde is an important branch of the economy and is one of the leading sectors in terms of exports (around 75 % of exports) ([1]). The activity is mostly based on fish and shellfish of high commercial value exported fresh, frozen and canned. In 2008 the sector employed 10,500 people, (61 % in the harvesting activity) ([2]).

Aquaculture

The waters of the archipelagos enjoy good conditions for the development of aquaculture, such as appropriate minimum and maximum temperatures throughout the year. This allows fish to reach commercial sizes before than in other regions. The activity started to develop in late 1990s. The Canary Islands is the leader in the region with a lager production seabass and seabream. Seabream is also produced in Madeira. The activity faces the problem of the insularity condition which substantially increases transport costs. EU's compensation for transport of these goods plays a key role in reducing transport costs.

Processing

Regarding processing, Azores has an important tuna canning activity. This is likely the largest cannery in the sea basin. In the past, an important canning industry existed in the Canary Island but, due to the end and renegotiation of access agreements to waters of third countries, the activity ended in early 2000s. The Madeira's canning industry also disappeared in early 2000s but for a different reason, the decline of tuna populations. There are processing activities in Cape Verde in hands of foreigner companies established in the archipelago they produce frozen and canned fish and shellfish. In relative terms, the Cape Verdean processing industry is the most important in the sea basin. This constitutes one of the most important sectors of the Cape Verdean economy.

1.1.2 Geographical distribution

The fishing and aquaculture activity takes place in the four archipelagos of the Macaronesia region: Azores, Madeira, Canary Islands, and Cape Verde.

([1]) Cabo Verde. Elaborado por la Oficina Económica y Comercial de España en Dakar. 2011. 40 % of the national exports are based on fish and shellfish. In turn, 30 % of the national exports are based on meat and fish canned. The country fiche 'Cabo Verde' published in July 2016 by the Ministry of Foreign Affairs of Spain states that fisheries account for 80 % of the Cape Verdean exports.

([2]) Coerência das Políticas para o Desenvolvimento no âmbito das pescas: Estudo de Caso em Cabo Verde. Damaris Rosabal Lopes da Silva — — Licenciada em Ciências do Mar, ULPGC; Mestre em Análise de Políticas para a Sustentabilidade, UPC. Cofinanciamento: União Europeia e Camões — — Instituto da Cooperação e da Língua, I.P. 2013.

1.1.3 Main socio-economic trends

The CAGR applied to GVA shows a negative trend of 7.18 % (period 2010-2013). In turn employment has maintained constant in the period 2010-2014 ([3]). In the same period, turnover increased (CAGR of 4.16).

1.1.4 Future prospects

In the Macaronesia sea basin, the fishing and aquaculture sector will face many issues for the coming years. The reduction of the fishing fleet implies a reduction of direct employment. In turn, it is reported that new generations do not have interest in the fishing activities, which will lead to ageing of crews. There may be different reasons for the problem of generational replacement. In the specific case of the ORs, this could be explained by the reduction of the fishing fleet, which does not seem to offer hope for the future. The activity in the OR is regarded also as an escape valve for workers in times of crisis. So, there is little incentive for youngsters to acquire formal qualifications. The wish of new generations to get jobs in growing urban areas may also have implications for the problems of generational replacement in the fishing activity. The sectors are highly dependent on quotas and agreements to access to third countries waters. This is a key driver for the development of the sector. In the case of bluefin tuna, for example, quota allocations is subject to three levels of decision-making: at the level of the ICCAT, at the level of the EU, and finally at the level of the Member States. The Canarian fleet is highly dependent on access to the waters of Western and Northern Africa countries. This has even determined the end of a large canning industry. So, the sector is strongly limited by external political factors. In terms of diversification, the Canary Islands are exploring the potential of deep sea fishing species such as black scabbard fish, which has a large market in Portugal. Aquaculture has good possibilities of diversification and the technical viability of production of corvina has been demonstrated by Canarian scientists. Algae production in turn has the potential to sustain a biotechnology sector. In the Canary Islands, the production is exported with little value added due to lack of processing facilities. Large investments would be required to integrate algae production with biotechnology in order to add value to this resource. Finally, the growth of the aquaculture industry face too long administrative process due to utilization of space and environmental standards.

1.2 Extraction of aggregates: operation of gravel and sand pits

This activity is conducted only in the Portuguese archipelagos of Azores and Madeira. In Azores, the activity consists of sand extraction, while in Madeira the activity comprises gravel and sand extraction. In both cases the activity has little economic weight and employs few people. This activity does not present good prospects due to the decline of demand and the high investments required to carry out the activity (see Portuguese ORs reports).

1.3 Desalination

This activity is carried out almost entirely in the Canary Islands. There is one plant in Madeira and seems sufficient to satisfy the local demand. The activity is being developed in Cape Verde. No official data are available on the number of plants being currently operated in the Cape Verdean archipelago ([4]). Development of the activity is receiving international support ([5]). In turn, the Canary Islands are amongst the world leaders in desalination of seawater. The large sector comprises 320 plants. This activity has been one of the factors allowing tourism development in the Spanish archipelago. This activity

([3]) Data for employment and turnover is not available for Cape Verde. Thus the CAGR is estimated only for the EU's ORs.
([4]) http://www.lavanguardia.com/economia/20150112/54422552524/acciona-agua-se-adjudica-dos-desaladoras-en-cabo-verde-por-9-6-millones.html.
([5]) http://www.macauhub.com.mo/en/2014/11/12/japan-pays-for-construction-of-desalination-plants-in-cabo-verde/.

consumes large amounts of electricity produced from fossil energy. This substantially increases costs, while producing an environmental impact. Experiences are being made in the Canarian archipelago to use on-land wind energy in the desalination plants.

1.3.1 Geographical distribution

The activity is predominately developed in the Canary Island. There is also one plant in Madeira.

1.3.2 Main socioeconomic trend

In the period 2010-2014, the activity decreased in terms of GVA and turnover in 1.84 % and 5.54 %, respectively. Employment in turn was maintained rather constant.

1.3.3 Future prospects

It seems that the growth of tourism and population in the region will increase demand of water and thus this industry will need to growth. The use of renewable energies and, especially, marine renewable energies has a great potential to solve the problem of high dependency on fossil energies (see the Canary Islands' OR report).

1.4 Maritime transport

1.4.1 Overview

The strategic position of the four archipelagos in the Atlantic Ocean confers advantages for the development of freight maritime transport. The atomization of the territory (the Canary Islands have 7 islands and Azores 9) make ferries the most important mean of transport of passengers in terms of number of passengers.

1.4.2 Geographical distribution

The activity is carried out particularly in the EU's archipelagos. Passenger transport amongst the islands is of high importance due to the problems imposed by the double insularity condition in the archipelagos.

1.4.3 Main socioeconomic trends

In the period 2010-2014 the CAGR shows a decrease in GVA and turnover of 0.39 % and 2.39 %, respectively. In turn employment grew in 1.85 %.

1.4.4 Future prospects

Some of the barriers to overcome in the inter-islands transport relate to the cost of transport and the difference in population density in the diverse islands of the archipelagos, which could make some of them more attractive and thus would have an impact on the offer of services. It is expected that growth of tourism and reactivation of the economic activity will require improved interisland connections. Azores and especially Canary Islands have a great potential for becoming logistic hubs in the Mid-Atlantic.

1.5 Ports

1.5.1 Overview

The largest and best equipped ports are located in Azores, Madeira and Canary Islands. Larges investments have been made to turn these ports into competitive infrastructures able to take advantage of the strategic location of the archipelagos. It is worth mentioning that the main ports such as Las Palmas, Santa Cruz and Ponta Delgada offer all services for freight ships, ferries and marinas. Other services offered in these ports are bunkering and ship repair. This is particularly relevant in the port of Las Palmas. A large array of secondary ports offers services to the fishing activity and nautical activities.

1.5.2 Geographical distribution

Ports and harbours are distributed all around the four archipelagos. However, the largest ports are Las Palmas, Santa Cruz and Ponta Delgada.

1.5.3 Main socioeconomic trends

In the period 2010-2014 the ports activity showed a slight decrease in GVA (0.39 %). Employment and turnover also decreased in (1.41 % and 0.85 %).

1.5.4 Future prospects

The strategic location of the Macaronesian ports in the Atlantic offer promising opportunities in terms of economic returns and employment in a diversity of activities directly and indirectly related to ports e.g. storing of grains and other goods, logistics for aid to African nations, bunkering for offshore fleets, nautical activities, cruise activity, etc. Shipbuilding has declined and its reactivation seems difficult. The ship repair activity offer opportunities for Canary Islands, especially in the field of fit and repair of recreational boats and services for the offshore fleets. Cooperation between the ports of the region will be required to establish a network of ports in the sea basin. Projects cofounded by the EU trough the INTERREG's Transnational cooperation program for Madeira, Azores and Canarias (MAC program) are being conducted with this aim (see section 3).

1.6 Shipbuilding and repair

1.6.1 Overview

Shipbuilding is a marginal activity in the Macaronesia sea basin. In Azores, the activity consists in construction of fishing vessels. In the Canary Islands, there is no an activity of steel vessels building. Wooden vessels are still been constructed but the activity is disappearing. Ship repair is also in decline in Azores and consist in repair of fishing vessels and recreational boats. Ship repair in the Canary Islands declined with the problem of access to third waters fishing grounds. Nowadays, however, the activities of inspection and maintenance of oil platforms and services to offshore fleets have contributed to the reactivation of the largest shipyards. This service is provided to platforms operating in the Gulf of Guinea (see the Canary Island's case).

1.6.2 Geographical distribution

These activities are carried out in Azores (Pico Island) and in the Canary Islands in diverse ports of the archipelago. The largest ship repair activities, however, are carried out in the larger ports of Las Palmas and Santa Cruz, where the inspection and maintenance of oil platforms is emerging as a significant economic opportunity.

1.6.3 Main socioeconomic trends

In the period 2010-2013 the activity's GVA experienced a substantial growth (16.33 %). Employment and turnover increased in 3.51 % and 17.64 %. This positive result is mostly due to the growth of the oil platform inspection and repair activity in the Canary Islands.

1.6.4 Future prospects

An activity showing good prospects is repair of recreational boats which require specialized labour. The activities of services to offshore fleets and oil platforms seems only relevant for the Canary Islands. Oil platforms must undertake comprehensive revisions every 5 years. The activity requires specialized abilities and formal qualifications. It is worth pointing out that the oil platform activity is subject to the fluctuations of oil price.

1.7 Coastal and cruise tourism

1.7.1 Overview

Tourism is a well-established activity in the Canary Islands and Madeira, where the activity accounts for a high share of GDP and employment, and is consequently one of the pillars of the economy in both archipelagos. The activity is backed up by large accommodation offer, air and maritime transport facilities, infrastructure for nautical sports, sport fields, amongst others. In contrast, the activity is quite recent in Azores and Cape Verde. In the case of coastal tourism, it shows great potential all around the Macaronesia sea basin. The old touristic model of sun and beach is gradually giving way to a new model of tourism that takes more advantage of the marine environment and its resources and of the coastal culture. In Cape Verde tourism, and especially coastal tourism, is gaining importance and in fact accounts for 50 % of the services sector, which in turn represents 60 % of the economy of this archipelago. Cape Verde is receiving considerable attention due to their natural appeal but also to the political stability and security. International investments in hotel infrastructure are being conducted in the archipelago. Nautical activities in Cape Verde represent opportunities for economic growth and for cooperation in the entire sea basin. In fact, progress have been made in the framework of the program MAC to conduct a diagnosis of the potential of nautical activities in that archipelago. It is worth mentioning that the transit of people between Cape Verde and the EU ORs requires a visa. This fact may limit mobilization of people between these regions. This has implications for tourism and other activities such as training of Cape Verde workers. In relation to cruise tourism the EU's ORs are both destination and repositioning places in the cruise routes. Madeira and the Canary Islands have made important efforts to improve infrastructures and the offer of packages to attract cruise tourists.

1.7.2 Geographical distribution

Coastal tourism and cruise tourism is carried out in the four archipelagos of the Macaronesia. Madeira and the Canary Islands are the most developed in cruise tourism.

1.7.3 Main socioeconomic trends

In the period 2008-2013, the GVA of tourism grew in 2.63 %. In relation to employment and turnover the increases were of 1.2 % and 2.08 %, respectively. It seems that the activity of tourism growths at a moderated but sustained pace.

1.7.4 Future prospects

New modalities of tourism, which take advantage of the marine environment and the cultural wealth of the coastal communities, are promising options to diversify the touristic offer. These alternatives include nautical activities, whale and bird watching, pesca-tourism, and marine gastronomy, amongst others. In relation to cruise tourism improving the offer of activities for cruisers would help strengthening the archipelagos cruise destinations.

It is worth mentioning that various projects are being conducted in the frameworks of projects funded by the EU's programme MAC to strengthen the competitiveness of Cape Verde's touristic sector, with special regards to nautical activities (see section 3).

1.8 Possible impacts of climate change in the Macaronesia sea basin ([6])

Under an environmental point of view, the archipelagos are facing an increase in temperature, a rise in sea level and reduction of the average annual rainfall. Concentration of rainfalls in certain periods might cause landslides and sea storms, in combination with the sea level rise, might cause floods. This threatens coastal areas and

([6]) European Commission, 2014, The economic impact of climate change and adaptation in the Outermost Regions, http://ec.europa.eu/regional_policy/sources/activity/outermost/doc/impact_climate_change_en.pdf .

infrastructures of vital importance for tourism, which is the main branch of the economy for the OR and an emerging activity for Cape Verde. Ports and maritime transport might also be affected posing a threat to blue growth in the area. Marine biodiversity, one of the main assets of the archipelagos, might be threatened for the rise in temperature and in sea level.

The fishing activity is also particularly vulnerable by the rise on sea temperature. This might imply changes in fishing patters with disappearance of traditional target species and arrival of tropical species. The fishing industry in Cape Verde might suffer more from climate change. Adaptive actions are being considered and priorities should be established for coastal protection, water, energy, amongst others. Cooperation with Cape Verde is also being formalised ([7]).

([7]) http://www.lavanguardia.com/local/canarias/20161121/412041021748/lanzarote-y-cabo-verde-aliados-cambio-climatico-y-sostenimiento-medioambiente.html.

2 Synergies and tensions between maritime activities

The table below reports synergies and tensions of each activity with other maritime activities and which are the contribution to sustainable Blue growth.

Group	Sector	Synergies with other maritime activities	Conflicts with other maritime activities	Contribution to sustainable growth
Living resources	Fisheries and aquaculture	In the Macaronesia region fishing is still an important driver of the development of coastal communities. It provides employment and contributes to food security. Fisheries can interact well with new modalities of tourism such as pesca-tourism and cultural tourism.	There is no evidence of conflicts with other activities at the level of the entire Macaronesia sea basin. At the level of the archipelagos the professional activities collide with recreational activities. This requires strict regulation and planning of the use of the marine space. The entry of fish captured by recreational fishers also represents an unfair competition with the professional sector.	Fisheries and aquaculture are relevant for sustainable growth provided that the activity is conducted in a sustainable way, respecting management measures and environmental considerations. Aquaculture shows great potential and is in fact in the light spot of the blue growth strategy. Sound research is being conducted especially in the EU's ORs in relation to aquaculture.
	Blue biotechnology	There is no a marine biotechnology activity as such in the sea basin. However, algae production in Madeira and Canary Islands has a great potential to provide a basis for a biotechnology activity directed to the food, cosmetic and pharmacy sectors.	Not applicable at the level of the sea basin.	It may substantially contribute to value addition of algae production.
	Extraction of aggregates	This activity is only relevant to the Portuguese ORs (see OR report).	Not applicable.	Not applicable.
	Desalination	This activity is not conducted at the whole sea basin level: only at the level of Madeira and the Canary Islands. In the Canary Islands desalination has had positive relations with tourism. Interaction with renewable energies shows also good perspectives.	Not applicable.	Water supply for the growth of economic activities and population substantially contributes to sustainable growth. In Madeira, the production seems sufficient. In the Canary Islands production increase is envisaged. Means have to be found to reduce consumption of fossil energy.
Shipping	Maritime transport	It is an element of integration in the Macaronesia region. It is of vital importance for economic development due to the insularity condition of the regions concerned.	The density of maritime routes will increase in the near future it will require effective coordination amongst the regions concerned.	The strategic position of the archipelagos makes maritime transport a vehicle of economic development and cooperation. Strengthening connections with Africa and South America will

Group	Sector	Synergies with other maritime activities	Conflicts with other maritime activities	Contribution to sustainable growth
				benefit all concerned nations.
	Ports (including dredging)	Ports are essential to sustain the maritime activities. Networks of port of the Macaronesia region are cooperating in the context of projects of the MAC program to improve coordination and knowledge transfer.	Not applicable.	Ports are essential for sustainable growth of marine and maritime activities. In fact, many activities linked to the blue economy are linked to ports e.g. fishing, cruise tourism, ship repair, etc.
Shipbuilding	Shipbuilding	This is only relevant at OR level (e.g. Canary Islands).	Not applicable.	Not applicable.
	Ship repair	This is only relevant at regional level (e.g. Canary Islands).	Not applicable.	Not applicable.
Tourism	Coastal tourism	This is only relevant at regional level. However, cooperation of the EU' ORs with Cape Verde is of key importance for its development. In fact, steps are being made to build this cooperation.	Not applicable.	Not applicable.
	Cruise tourism	This is mostly relevant at regional level. Nevertheless, cooperation in establishing routes and services could be conducted to attract more cruises to the Macaronesia sea basin.	See maritime transport, above.	Not applicable.
Marine renewable energies	Wind energy	This is mostly relevant at regional level.	Not applicable.	Not applicable.

2.1 Synoptic view of legislative, non-legislative and financial synergies and tensions among maritime activities

The following table reports a synoptic view of main synergies and tensions grouping them under legislative, non-legislative and financial frames.

Synergies with other maritime activities			
ACTIVITY	LEGISLATIVE	NON-LEGISLATIVE	FINANCIAL
Fisheries and aquaculture	Synergies have been identified between the fishing sectors of the ORs in the framework of the South Western Waters Advisory Council (SWW AC). Conversations are being held in relation to the potential AC of the ORs.	In the sea basin, fisheries can interact well with new modalities of tourism such as pesca-tourism and cultural tourism.	• EMFF is the key instrument for the development of the fishing and aquaculture activity in the ORs of the sea basin. • The program MAC is a key driver for R&D and knowledge transference in the four Macaronesia's archipelagos.
Blue biotechnology	No synergies are identified.	There is no a marine biotechnology activity as such in the sea basin. However, algae production in Madeira and Canary Islands has a great potential to provide a basis for a biotechnology activity directed to the food, cosmetic and pharmacy sectors.	No synergies are identified.
Extraction of aggregates	Only relevant for the Portuguese ORs. No synergies are identified at sea basin level.		
Desalination	No synergies are identified.	The activity is carried out only in Canarias and Madeira. It has strong implications for tourism. Effluents of the desalination process are useful for microalgae production. Desalination will have a key importance for Cape Verde due to the growth of tourism.	No synergies are identified.
Maritime transport	No synergies are identified.	It is of vital importance for economic development of the sea basin due to the insularity and double insularity condition of the regions concerned.	No synergies are identified.
Ports (including dredging)	No synergies are identified.	Networks of ports of the Macaronesia region are cooperating in the context of projects of the MAC program to improve coordination and knowledge transfer. This activity is of high relevance for maritime transport, cruise tourism and ship repair.	No synergies are identified.
Shipbuilding	No synergies are identified.	This is only relevant at OR level (e.g. Canary Islands and Azores).	No synergies are identified.
Ship repair	No synergies are identified.	This is only relevant at OR level (e.g. Canary Islands and Azores). These OR have the potential to provide services to the maritime sectors of the sea basin e.g. shipping,	No synergies are identified.

Synergies with other maritime activities			
ACTIVITY	LEGISLATIVE	NON-LEGISLATIVE	FINANCIAL
		nautical activities.	
Coastal tourism	No synergies are identified.	Cooperation of the EU' ORs with Cape Verde is of key importance for its development. In fact, steps are being made to build this cooperation.	No synergies are identified.
Cruise tourism	No synergies are identified.	Cooperation is being conducted to attract more cruises to the Macaronesia sea basin in the framework of private initiatives.	No synergies are identified.
Wind energy	No synergies are identified.	This is mostly relevant at regional level and carried out on land, e.g. in Canarias.	No synergies are identified.

Conflicts with other maritime activities			
ACTIVITY	LEGISLATIVE	NON-LEGISLATIVE	FINANCIAL
Fisheries and aquaculture	No conflicts are identified.	There are not evident conflicts with other activities at the level of the entire Macaronesia sea basin. Conflicts between fisheries and aquaculture and other uses of the coastal areas are reported in the OR reports. The entry of fish captured by recreational fishers also represents an unfair competition with the professional sector.	No conflicts are identified.
Blue biotechnology	No conflicts are identified.		
Extraction of aggregates	No conflicts are identified.		
Desalination	No conflicts are identified.		
Maritime transport	No conflicts are identified.	The density of maritime routes will increase in the near future it will require effective coordination amongst the regions concerned.	No conflicts are identified.
Ports (including dredging)	No conflicts are identified.		
Shipbuilding	No conflicts are identified.		
Ship repair	No conflicts are identified.		
Coastal tourism	No conflicts are identified.		
Cruise tourism	No conflicts are identified.		
Wind energy	No conflicts are identified.		

3 Best practices and flagship projects

1. **Name of the project/ practice**: MARES
   Objectives:
   - creation and development of a virtual entity; coordination and identification of a common strategy for R & D and innovation;
   - defining a plan of R & D and innovation common to the three EU Macaronesian regions, to strengthen research, technology, and innovation and information society;
   - promoting mobility of students and researchers between regions in the Macaronesia, reinforcing a growing collaboration between researchers
   - creating a common living Lab environment for research and innovation in the areas identified.

   Results achieved: The project has strengthened links amongst the three EU' archipelagos in relation to R&D in the context of marine and maritime activities, which provides a basis for the blue growth in the sea basin.
   Reasons for considering it as flagship project: It develops cooperation in R&D and education at sea basin level in order to provide a basis for economic growth.

2. **Name of the project/ practice**: ALGABIOMAC.
   Objectives: The main objective is the development of activities of exploitation, production, processing and promotion of seaweed (macro and micro) in Macaronesia. Cooperation actions and strategies in research, technological development that enable the viability and exploitation of these resources, and the development of biotechnology companies.
   Results achieved: Assessment, determination and extraction of bioactive substances, both microalgae and macroalgae regions of Macaronesia; Evaluation and technical and commercial characterization of supercritical extraction of natural pigments, antioxidants (carotenes) and fatty acids (omega 3 and 6) of microalgae and macroalgae for the development of new industries; Study of technical and commercial viability of the production of micro and microalgae in Macaronesia; Establishment of research networking in the region and training of specialists.
   Reasons for considering it as flagship project:
   The project addresses one of the main needs in the field of Blue Growth which is to determine the technical and commercial viability of the extraction of bioactive substances. This activity has the potential to add value to the current microalgae production. The projects promote scientific and technical cooperation at the level of the entire sea basin but similar initiatives could be extended to the other ORs.

3. **Name of the project/ practice**: BIOVAL.
   Objectives: The main objective is to undertake cooperation in research and technological development using molecular biology techniques, especially those related to DNA, to promote R & D and innovation in the Macaronesian region and promote its development, through its application to marine resources.
   Results achieved: Results are not available.
   Reasons for considering it as flagship project: The project is of particular interest since it improves scientific and technical cooperation in marine sciences for the regions in the Macaronesia sea basin.

4. **Name of the project/ practice**: ORFISH.
   Objectives: Development of innovative, low-impact offshore fishing practices for small-scale vessels in outermost regions.
   Results achieved: In progress.
   Reasons for considering it as flagship project: This project is highly relevant since deals with small scale fisheries (important activity for all ORs).

5. **Name of the project/ practice**: GESPORT.
   Objectives: To create a Euro-African network for sustainable cooperation in the port sector, enhancing in this way the territorial cooperation in line with the European Union cohesion policy.
   Results achieved: Not available.
   Reasons for considering it as flagship project: This project is relevant not only for the Macaronesian sea basin but also for other ports in Africa.

6. **Name of the project/ practice**: ISHLAGUA.
   Objectives: The aims of the project were to raise public awareness about efficient water consumption, strengthen quality control capacity of water resources, develop the treatment and reuse of treated water, and improve water desalination efficiency together with increasing.
   Results achieved: Institutional capacity building in water depuration, and reutilization of residual waters; desalination of seawaters using renewable energies.
   Reasons for considering it as flagship project: The project strengthens cooperation between the Canary Islands and Cape Verde in the development of the desalination sector, amongst others.

7. **Name of the project/ practice**: MAC PROF-CV.
   Objectives: To evaluate potential new deep-water resources in Cape Verde, and to provide the bases for sustainable management, and gastronomic valuation.
   Results achieved: A deep-sea evaluation survey was conducted on the abundance of striped soldier shrimp and black scabbard fish in Cape Verde waters. In addition, the project conducted culinary events to promote the consumption of these and other fisheries resources.
   Reasons for considering it as flagship project: This project explored potential resources to diversify the fishing activity and promote further development of fisheries in Cape Verde. This initiative could be replicable in other contexts.

8. **Name of the project/ practice**: CANAUTIC.
   Objectives: The platform for 'Cooperation in nautical sports Canary Islands -Cape Verde, socioeconomic and environmental diagnosis in relation to nautical activities' is aimed to promote sustainable territorial development of coastal areas of Cape Verde, while establishing a framework for the rational use of the coastal zone based on social, economic and environmental criteria.
   Results achieved: Socio-economic diagnoses of the nautical sector of Cape Verde. Catalogue of the nautical activities of the Cape Verde and advice for the development of the regulatory framework for the nautical activity.
   Reasons for considering it as flagship project: Provides a diagnosis of the potential of the Cape Verde nautical sector and contributes to its institutional development. The project is a breakthrough in evaluating the potential the Cape Verde nautical

sector and does a meaningful contribution to its institutional development. The project could be replicable to other regions where the nautical activity is incipient since it is aimed not only at identifying potentials but also at proposing regulatory frameworks.

9. **Name of the project/ practice**: ESTRAMAR
   Objectives: To promote R&D and innovation to contribute to better coordination of the scientific-technical enterprise system, in fields such as security and sustainable transport, maritime tourism and ports, increased protection of coastal areas, resources and marine biodiversity and the provision and management of natural risks, and thereby contribute to boosting socio-economic development of these regions. This objective is a specific contribution to improve the system of R&D and innovation, where the Macaronesian regions have specific conditions and potentials.
   Results achieved: Results are not available.
   Reasons for considering it as flagship project: The project is of particular interest since it aims at making a better use of the marine space for the diverse marine and maritime activities, while preserving the marine environment and its resources.

10. **Name of the project/ practice**: PTMAC
    Objectives: The Platform is a public-private structure formed by a network of enterprises, NGOs, research groups and other entities aiming at promoting innovation and development of the economic sectors of the archipelagos of Macaronesia.
    Results achieved: Constitution of the platform and drafting of its agenda. Working groups were launched in the strategic sectors of agriculture-food and nautical tourism. A new application for MAC support is to be carried out.
    Reasons for considering it as flagship project: This participatory approach has a great potential to contribute to economic development by gathering the insights and experience of the diverse actors concerned, which enable more informed strategies.

11. **Name of the project/ practice:** Grupo de Investigación en Acuicultura.
    Objectives: The group promotes a strategic vision of the aquaculture sector with products of high nutritional quality, innovation with sustainable diets, selection of species and multitrophic polyculture, encouraging a model of social and biotechnological development based on local species.
    Results achieved: The group has developed and registered eight services, which comprise analysis of nutritional value aquatic organisms; models and genetic tools; histopathologic diagnosis; production of eggs, larvae and juveniles of diverse species; sensorial analysis of aquatic products; formulation and elaboration of feed; design and execution of aquaculture essays, amongst others. These services are offered to researchers, research and academic institutes and private sector.
    Reasons for considering flagship project: The group is an innovative good practice that capitalize the scientific knowledge, know how on productive processes and state of the art infrastructure to provide a wide array of products and services to the private sector and to researchers and other research institutions. Thus, it does not only back up industry with scientific expertise but also build capacity in the research and development realm. The model of the research group could be replicable in other contexts provided that scientific expertise and adequate infrastructures are in place.

3.1 Best practices and flagship projects details

Name of the project/ practice	Leading actor(s)	Other subjects/ stakeholders involved	Timing	Maritime sectors involved	Budget	Programming document	Website
MARES	PLOCAN	TECNOPOLO (Madeira) Regional Goverment of Azores	2007-2013	R&D	549,724 € (467,266 € ERDF)	MAC 2007-2013	http://www.plocan.eu/index.php/en/portfolio-proyectos/1196
ALGABIOMAC	The Canarian Institute of Cancer Research (FICIC)	Instituto Tecnológico de Canarias (ITC); Scientific Associação de Investigação do Atlântico (AICA); Câmara Municipal Ilha do Maio, Cape Verde	10/2013 — 10/2015	Biotechnology, aquaculture	240,650 €	MAC 2007-2013 (partial funding— 192 000.56)	http://www.algabiomac.com/
BIOVAL	ULPGC	Instituto Canario de Ciencias Marinas; (ITC); Câmara Municipal do Funchal (CMF); Museu Municipal do Funchal (História Natural) (MMF); Estação de Biologia Marinha do Funchal (EBMF); Universidade da Madeira; Universidade de Cabo Verde; Universidade dos Açores	07/2013	09/2015	241,509.70 €	MAC 2007-2013 (partial funding — 192.000)	http://www.pct-mac.org/registroficha?id=90d733da-29be-4a57-8641-892f9f6b00bc
ORFISH	Représentation de la Région Guadeloupe à Bruxelles	Universidad de La Laguna, Instituto Español de Ocenaografía (IEO) — Canarias, Centro de Investigaciones Ambientales CIMA Canarias, Gestión del Medio Rural de Canarias, Guadeloupe Region, CRPMEM Guadeloupe, Carbet des sciences-CCSTI (Martinica), Sea expert, serviços e consultoria na area das pescas, Lda (Azores), Federaçao das pescas dos Açores, Regional Fisheries Dept of GOV. of MADEIRA, Coopescamadeira, Universidade de Aveiro (Portugal), CAPAM Mayotte, CRPMEM Réunion, IFREMER, Universidad de Brest (Francia), Wemake SME (Francia).	08/2016 — 12/2017	Offshore fisheries	999,511€	DG MARE. Call for Proposals MARE/2015/06	http://www.azores.gov.pt/Portal/en/entidades/srmct-drp/noticias/Government_of_the_Azores_included_in_European_experimental_project_on_fisheries.htm
GESPORT ([8])	Port Authority of Santa Cruz de Tenerife	Port Authority of Las Palmas, their counterparts in the ports of Madeira, Gambia and Cape Verde, and the autonomous ports of Dakar, Nuadib and Nouakchott	2009-2013	Shipping (Port)	453,750 €	European transnational cooperation project (MAC 2007-2013)	http://www.puertosdetenerife.org/memorias/Memoria2011/7/Chapter-07-port-

([8]) http://www.puertosdetenerife.org/memorias/Memoria2011/7/7-port-activity-en.html.

Name of the project/ practice	Leading actor(s)	Other subjects/ stakeholders involved	Timing	Maritime sectors involved	Budget	Programming document	Website
							activity.pdf
ISLHAGUA	Technological Institute of the Canary Islands (ITC)	Mancomunidad del Sureste de Gran Canaria, Association of towns of Cape Verde, University of Cape Verde, National Institute of Hydrological Resources of Cape Verde	From 11/2010 to 06/2014	Water resources	463, 680 €	European transnational cooperation project (MAC 2007-2013)	http://www.islhagua.org/web/guest;jsessionid=7912D1BD25A9047F4F41EA54339787F2
MARPROF-CV	Instituto Canario de Ciencias Marinas (ICCM)	Direcção de Serviços de Investigação das Pescas (SRA/DRP/DSIP) Madeira Câmara Municipal do Funchal / Museu Municipal do Funchal Madeira Instituto Nacional de Desenvolvimento das Pescas (INDP), República de Cabo Verde	10/2010 12/2015	Fisheries, R+D and innovation, Tourism	671,842 € 543,614 € (ERDF)	MAC 2007-2013	http://www.pct-mac.org/registroficha?id=733cb2f7-90fd-458b-aed0-bc41fe371950
CANAUTIC	Centro Tecnológico de Ciencias Marinas (CETECIMA)	PLOCAN Universidad de Las Palmas de Gran Canaria—ULPGC Instituto Marítimo e Portuário—IMP (Cape Verde)	09/2013-11/2015	Tourism, Nautical activities	160,000€ (ERDF)	MAC 2007-2013	http://canautic.com/
ESTRAMAR	Universidad de Las Palmas de Gran Canaria (ULPGC)	Universidad de Las Palmas de Gran Canaria (ULPGC) Canarias, Instituto Canario de Ciencias Marinas Canarias, Universidade dos Açores Azores, INDP, Instituto Nacional De Desenvolvimento Das Pescas Cabo Verde, PLOCAN, Fundação Gaspar Frutuoso	10/2010-10/2015	Maritime transport, ports, maritime tourism, amongst other sectors	648,056€ 523,775 €(ERDF)	MAC 2007—2013	http://estramar.eu/
PTMAC	Universidad de La Laguna	Universidad de La Laguna (ULL), Universidade da Madeira, Universidade das Azores, Cluster do Mar (Cape Verde), FEMETE, Cluster Marítimo de Canarias, Arditi (Agência Regional para o Desenvolvimento da Investigação Tecnologia e Inovação), and Fundación Gaspar Fructuoso.	06/2014-12/2015	Nautical tourism, among others non-maritime sectors	214,099€ 172,974€ (ERDF)	MAC 2007-2013	https://ptmac.webs.ull.es/

4 Gap analysis

In this section, each maritime activity has been analysed in order to identify 'what is missing for a sustainable Blue Growth' in terms of 'Human resources skills', 'Infrastructures', 'Research' and 'Data'.

- ***Fisheries and aquaculture***

Human resources skills	Fishing is a traditional activity in the Macaronesia region. Crews are aged and often do not hold appropriate educational qualifications. In relation to aquaculture the activity is mostly developed in the Canary Islands, where technical training is available. The other regions have limitations in terms of training infrastructure.
Infrastructures	IUU fishing is pointed out as a problem and requires an effective monitoring, control and surveillance system. Potential species of economic interest, especially deep-water species and small pelagic species, deserve particular attention in order to evaluate their potential. Technical developments to catch deep water species are also required. Market studies are required to assess the viability of the introduction of new species in the regional market and other markets. Production of algae is conducted in Madeira and Canary Islands. Infrastructures to add value to the production trough application of biotechnology are required to take advantage of market niches i.e. cosmetics, food, pharmacy. Another gap for the growth of aquaculture production is the difficulty to access to treatment of fish diseases due to the cost of importing medicines, time of transport and custom procedures. This is related with the insularity and remoteness of the region ([9]).
Research	In Cape Verde, R&D in fisheries and marine matters is limited.
Data	There is a lack of data for Cape Verde fisheries.

- ***Desalination***

Human resources skills	In Cape Verde, the growth of population and touristic activity requires a larger desalination activity. This requires personnel with medium and high level of qualification. Specialized training is even scarce in the Canary Islands where postgraduate studies are not available for engineers.
Infrastructures	Cape Verde lacks the desalination capacity to address increasing water needs. Efforts are being made by developed nations to support the development of this activity.
Research	Although R&D in this activity is active in the Canary Islands there is still room for improvement in water quality and utilisation of alternative energy sources in the plants.
Data	There is a lack of data for Cape Verde. In the Canary Islands socioeconomic data is offered aggregated with other water activities.

([9]) http://www.ipacuicultura.com/noticias/en_portada/33177/la_sanidad_acuicola_como_factor_clave_para_la_supervivencia_y_desarrollo_del_sector.html.

- ***Maritime transport***

Human resources skills	In Portuguese and Spanish ORs training of human resources need to be improved, especially in shipping and logistics management. This is a need to face the growing demand of maritime transport services.
Infrastructures	Support services linked to ships, freight and crews; and more frequencies to small islands are required in Portuguese ORs.
Research	-
Data	In general data on passenger water transport are incomplete. In the Canary Islands there is more complete information on passengers and freight.

- ***Ports***

Human resources skills	In Portuguese and Spanish ORs training of human resources need to be improved, especially in relation to the emerging offshore activities. This is a need to face the growing demand.
Infrastructures	In the Portuguese ORs Improvements on support services linked to ships, freight and crews are needed. Equipment and supplies for the ship repair industry are also required. In the Canary Islands improvements are needed for facing the increasing demand of bunkering services. Facilities for nautical activities need to be expanded to face the growing demand.
Research	-
Data	In general incomplete/lacking economic data on activities incidental to water transportation and nautical activities.

- ***Shipbuilding and repair***

Human resources skills	Shipbuilding is, nowadays, a marginal activity in Azores and the Canary Islands. Limited skills for ship repair e.g. utilisation of synthetics materials for hull repair are reported in the whole sea basin.
Infrastructures	Limited space in ports for ship repair is reported, especially in the Portuguese ORs. Limited provision of equipment and material is also reported.
Research	-
Data	In general incomplete/lacking economic data ship building and repair activities.

- ***Coastal tourism***

Human resources skills	Qualification of the human resources and training is required. In many cases, the activity has developed in areas where a low level of education predominates. Training activities are being conducted to respond to punctual demands but there seem a lack of a steady offer of training. The seasonal pattern of the activity may have had implications in the lack of qualification and professionalization in the activity.
Infrastructures	In many places, accommodation facilitates seems obsolete, especially on those areas with many decades of touristic activity. Nautical infrastructure or expansion of the existing ones is needed to satisfy the increasing demand.
Research	-
Data	Lack of data on nautical activities in terms of economic variables and people employed, which impedes knowing the economic relevance of this promising activity.

- ***Cruise tourism***

Human resources skills	There is a reported lack of training offer in Madeira and Cape Verde.
Infrastructures	There is a reported lack of space in the Portuguese ORs ports although efforts are being conducted to improve infrastructure. Limited satisfactory conditions for cruise stopovers in the diverse ports of the sea basin.
Research	N/A
Data	Specific data about this activity lacks in the entire sea basin. Data on EU's archipelagos provided in this study have been estimated using the consultants' methodology.

4.1 Synoptic view on the gaps identified

The following table reports a synoptic view of main gaps identified grouping them by 'legislative', 'non-legislative' and 'financial' gaps.

ACTIVITY	LEGISLATIVE	NON-LEGISLATIVE	FINANCIAL
Fisheries and aquaculture	An effective monitoring, control and surveillance apparatus is required to conduct surveillance of the fishing activity in the waters of the sea basin in order to deter IUU fishing.	• Crews are aged and, in many cases, do not hold appropriate professional qualifications. Technical developments to catch deep-water species are required. • ORs fleets are aged and lack security on board e.g. Madeira. • Ports for small scale fishing activities lack ice capacities e.g. in the Canary Islands. • Transport costs are high due to remoteness, insularity and double insularity. • Development of aquaculture in regions such as Cap Verde will face constrains on personnel with technical qualifications. • Production of algae is conducted in Madeira and Canary Islands. Infrastructures to add value to the production trough application of biotechnology are not available to take advantage of market niches i.e. cosmetics, food, pharmacy. • There is a lack of data for Cape Verde fisheries. • The growth of aquaculture production is facing difficulty to access to treatment of fish diseases due to the cost of importing medicines, time of transport and custom procedures.	Private investment at the level of the OR is scarce. There is a high dependency on structural funds. National public investment is scarce for the development of the fisheries sector in Cape Verde, which relies heavily on FDI.
Desalination		• Cape Verde still lacks the capacity to desalinate water for the increasing demand. • There is a lack of a specialized medium and high level training in Cape Verde and even in the Canary Islands.	
Maritime transport		• In the ORs training of human resources need to be increased to respond to the new challenges of the growth of the activity, especially in shipping and logistics management.	
Ports		• In Portuguese and Spanish ORs training of human resources need to be improved, especially in relation to the emerging offshore activities. • In the Portuguese ORs Improvements on support services linked to ships, freight and crews are needed. Equipment and supplies for the ship repair industry are also required. • In the Canary Islands improvements are needed for facing the increasing demand of bunkering services. Facilities for nautical activities need to be expanded to face the growing demand.	There is little investment in the Cape Verde ports. Investments in the regions are made at the level of the ORs, employing ERDF.
Shipbuilding and repair		• Limited skills for ship repair e.g. utilisation of synthetics materials for hull repair are reported in the whole sea basin. • Limited space in ports for ship repair and provision of equipment and material are reported, especially in the Portuguese ORs.	
Coastal tourism		• Qualification of the human resources and training is required. Training activities are being conducted to respond to punctual demands but there seems to be a lack of a steady offer of training.	
Cruise tourism		• There is a reported lack of space in the Portuguese ORs ports although efforts are being conducted to improve infrastructure. • Limited satisfactory conditions for cruise stopovers in the diverse ports of the sea basin. • There is limited training offer, especially in Madeira and Cape Verde.	

Mapping actors and maritime strategies

5 Mapping public authorities

Table 4 — Public authorities surveyed in Macaronesia

OR/sea basin ([10])	Maritime sector(s)	Name of the authority	Main roles and responsibilities	Address	Website
Cape Verde	Fisheries and aquaculture	Ministry of Agriculture and Environment	This ministry is responsible for the development and regulation of environment and primary industries in Cape Verde.	Palácio do Governo—Várzea Cidade da Praia Ilha de Santiago 304 Cabo Verde	http://www.governo.cv/index.php/component/content/article/81-conteudos-v/geral/81-ministro-da-agricultura-e-ambiente
Cape Verde	All maritime sectors	Ministry of Economy and Employment	This ministry is responsible for the development and regulation of economy and employment in Cape Verde	Palácio do Governo—Várzea Cidade da Praia Ilha de Santiago 304 Cabo Verde	http://www.governo.cv/index.php/component/content/article/81-conteudos-v/geral/87-ministro-da-economia-e-emprego
Canary Islands	R&D, innovation, capacity building for all productive sectors, including the marine and maritime sectors	Agencia Canarias de Innovación y Sociedad de la Información (ACIISI)	The Canarian Agency for Research, Innovation and Information Society (ACIISI), under the overall direction of the head of the Ministry of Economy, Industry, Trade and Knowledge, is responsible for implementing the public policies and programs in research, technological development, business innovation and deployment of the information society of the Autonomous Community of the Canary Islands. This agency is in charge of the implementation of the RIS3 strategies in the Canary Islands, where the Blue Growth strategy is framed. The fields of competence of ACIISI are: Research; Innovation; Society of Information; Development of human capabilities and innovative research; scientific and technological infrastructures; Support of the Coordination Commission of Science, Technology and Innovation	C/ León y Castillo, nº 200 Edf. Servicios Múltiples III 6ª 35071 Las Palmas de Gran Canaria Avda. Buenos Aires, nº 5 Edf. Tres de Mayo Planta 2ª 38071 Santa Cruz de Tenerife	http://www.gobiernodecanarias.org/aciisi/la_agencia/index.html
Canary Islands	Tourism	Consejería de Turismo, Cultura y Deportes	In charge of planning and promotion of tourism in the Canary Islands	C/ León y Castillo, nº 200 Edf. Servicios Múltiples III 6ª 35071 Las Palmas de Gran Canaria	http://www.gobiernodecanarias.org/ctcd/
Canary	Fisheries and	Consejería	The Consejería (Ministry) of Agriculture,	Avda. Alcalde José Ramírez	http://www.gobiernodecanarias.

([10]) Only authorities with direct relevance in promoting blue growth at the level of the Macaronesia sea basin.

OR/sea basin ([10])	Maritime sector(s)	Name of the authority	Main roles and responsibilities	Address	Website
Islands	aquaculture	de Agricultura, ganadería, pesca y aguas	Livestock, Fisheries and Water of the Canary Islands Government prepares and implements the governmental policy of agricultural, fisheries, food policy, and land surface and underground waters. In addition, it is responsible for implementing the Common Agricultural Policy (CAP) and the Common Fisheries Policy (CFP) in the Canary Islands. The Directorate General for Fisheries of the Consejería regulates interior sea fishing, shellfish and aquaculture waters; management, commercialization and industrialization of fishing, shellfish and aquaculture sector; fisheries inspection and monitoring; fisheries research and training both professional and recreational and underwater activities.	Bethencourt, nº 22 Edf. Jinámar Planta 2ª 35071 Las Palmas de Gran Canaria Avda. José Manuel Guimerá, nº 10 Edf. Servicios Múltiples II Planta 4ª 38071 Santa Cruz de Tenerife	org/agricultura/consejeria/
Canary Islands	Port	Port Authority of Las Palmas	Public institution that manages five ports: La Luz, Salinetas, Arinaga, Marbles and Puerto del Rosario; in 3 different islands: Gran Canaria, Lanzarote and Fuerteventura	Tomás Quevedo Ramírez, s/n 35008 Las Palmas de G.C. Las Palmas	http://www.palmasport.es/web/guest/autoridad-portuaria
Canary Islands	Port	Port Authority of Santa Cruz de Tenerife	Public institution that currently manages the ports of Santa Cruz de Tenerife, Santa Cruz de La Palma, Los Cristianos, San Sebastian de La Gomera and La Estaca	Avenida de Francisco La Roche, 49—Edif. Junta del Puerto 38001 Santa Cruz de Tenerife	http://www.puertosdetenerife.org/index.php/es/
Canary Islands	Port	Canary ports	Planning, operation and management of the port system owned by the Canary Island Government.	Plaza del Fuero Real de Gran Canaria, nº 4 Edf. Tamarco Planta Baja 35071 Las Palmas de Gran Canaria	https://puertoscanarios.es/
Autonomous Region of Madeira	Sea policy: safety, tourism, sport, knowledge and education, research and innovation, protection and conservation,	DROTA — Direção Regional do Ordenamento do Território e Ambiente	Ensure implementation of the policy defined by the Regional Government for the policy of the Sea	Rua Dr. Pestana Júnior 6º, 3º D 9064-506 Funchal	http://www.madeira.gov.pt/drota

OR/sea basin ([10])	Maritime sector(s)	Name of the authority	Main roles and responsibilities	Address	Website
	sustained and sustainable exploitation and the respective value of living and non-living resources				
Autonomous Region of Madeira	Fishing and aquaculture	Direção Regional de Pescas	Ensure implementation of the policy defined by the Regional Government for the fishing and aquaculture sectors	Edifício da Sociedade Metropolitana de C. de Lobos Praça da Autonomia 9300-138 Câmara de Lobos	http://www.madeira.gov.pt/srap
Autonomous Region of Madeira	all	Governo da Madeira — Direção Regional de Assuntos Europeus e Cooperação Externa	European affairs	Rua Câmara Pestana, nº 17-2º, 9000-043 Funchal	http://www.madeira.gov.pt/srape/Estrutura/Dire%C3%A7%C3%B5es-Regionais
Autonomous Region of Azores	All	Direção Regional dos Assuntos do Mar	All activities related to the sea	Rua Cônsul Dabney — Colónia Alemã Apartado 140 9901-014 Horta	http://www.azores.gov.pt/Portal/pt/entidades/srmct-dram/
Autonomous Region of Azores	Fishing and aquaculture	Direção Regional das Pescas	Ensure implementation of the policy defined by the Regional Government for the fishing sector	Rua Cônsul Dabney — Colónia Alemã Apartado 140 9901-014 Horta	http://www.azores.gov.pt/Portal/pt/entidades/srmct-drp/

6 Summary of existing maritime strategies

The strategies attempt to involve the four archipelagos of Macaronesia. Thus, these strategies are made at the level of the entire sea basin. Promotion of marine and maritime activities is sought trough cooperation and use of EU funds for regional development. Particular attention is paid to the needs of Cape Verde in developing its touristic, maritime transport and nautical capabilities.

6.1 Maritime Cluster of the Macaronesia

- **Geographical level (local/national/sea basin)**

Macaronesia sea basin.

- **Maritime sectors involved**

All maritime sectors, especially coastal and cruise tourism, nautical activities, aquaculture, renewable energies, naval activities, research, development and innovation, amongst others.

- **Objectives**

Its main objective is to promote growth and sustainable employment, according to the maritime strategy for the Atlantic.

- **Implementing tools**

Cooperation and development of synergies in the maritime sectors with the aim of proposing projects and make use of legal and financial instruments of the EU in the period 2014-2020. It is worth pointing out that the creation of the cluster was prompted by the project MARES (see section 3).

- **Actors involved and beneficiaries**

Members of the cluster are: Gobierno de Canarias, Ayuntamiento de Las Palmas de Gran Canaria, PLOCAN, Autoridad Portuaria de Las Palmas, Cámara de Comercio e Industria de Madeira, Cámara de Comercio de Gran Canaria, el Clúster Marítimo Español, el Clúster Marítimo de Canarias, la Federación de Empresas del Metal y Nuevas Tecnologías de Santa Cruz de Tenerife (FEMETE), el Clúster Empresarial de las Energías Renovables, el Medio Ambiente y los Recursos Hídricos de Canarias, el Clúster Canarias Excelencia Tecnológica, ASNAUTICA, ARN Y AENÁUTICA. The presidency is held by Cluster Marítimo de Canarias.

Beneficiaries are the whole maritime sector of the OR concerned and society as a whole.

- **Possible contribution to the fulfilment of BG objectives**

The coordinated activities of the cluster attempt to take advantage of the financial opportunities provided by the EU for the period 2014-2020.

- **Website**

http://plocan.eu/b3m/b3m-n8.pdf; http://www.clustermc.es/

6.2 Technological Platform of the Macaronesia

- **Geographical level (sea basin)**

Macaronesia region

- **Maritime sectors involved**

Agro-food sector, nautical activities, research, development and innovation

- **Objectives**

The Platform is a public-private structure formed by a network of enterprises, NGOs, research groups and other entities aiming at promoting innovation and development of

the economic sectors of the archipe-lagos of Macaronesia. The Platform is not a permanent structure it is organised as an EU funded R+D and innovation project (MAC program) and, as such, its activity is subject to the temporal frame of the project.

- **Implementing tools**

The platform establishes a framework for defining priorities in innovation and development. It creates working groups in the strategic sectors of agriculture-food and nautical tourism. It develops agenda for these two strategic sectors.

- **Actors involved and beneficiaries**

Universidad de La Laguna, Universidade da Madeira, Universidade das Azores, Cluster do Mar (Cape Verde), FEMETE, Cluster Marítimo de Canarias, Arditi (Agência Regional para o Desenvolvimento da Investigação Tecnologia e Inovação), and Fundación Gaspar Fructuoso. Beneficiaries are the entrepreneurs of the Macaronesia region and R+D and innovation agents.

- **Possible contribution to the fulfilment of BG objectives**

One of the strategic sectors of the platform is nautical tourism which has a great potential for employment and income generation in the Macaronesian sea region as a whole.

- **Website**

https://ptmac.webs.ull.es/

6.3 Marine-Maritime Strategy of the Macaronesia

- **Geographical level (local/national/sea basin)**

Macaronesia sea basin.

- **Maritime sectors involved**

Maritime transport, ports and Coastal tourism.

- **Objectives**

To promote the marine-maritime R + D and innovation of the European-African Macaronesian regions in order to contribute to better coordination of the scientific-technical enterprise system, in fields such as security and sustainable transport, maritime tourism and ports, increased protection of coastal areas, resources and marine biodiversity and the provision and management of natural risks, and thereby contribute to boosting socio-economic development of these regions.

- **Implementing tools**

To develop an oceanographic observation system with particular focus on Cape Verde.

- **Actors involved and beneficiaries**

Universidad de Las Palmas de Gran Canaria (ULPGC) Canarias, Instituto Canario de Ciencias Marinas Canarias, Universidade dos Açores Azores, INDP, Instituto Nacional De Desenvolvimento Das Pescas Cabo Verde, PLOCAN and Fundação Gaspar Frutuoso.

- **Possible contribution to the fulfilment of BG objectives**

To improve the quantity and the quality of the data of oceanic observation and promote good environmental practices.

- **Website**

http://www.estramar.eu/

6.4 Association cruises in the Atlantic

- **Geographical level (local/national/sea basin)**

Macaronesia sea basin.

- **Maritime sectors involved**

Cruise tourism.

- **Objectives**

The association conducts studies and surveys to improve port facilities and to know cruise passengers and crew latest needs.

- **Implementing tools**

Promotion of the destinations trough surveys and participation in events.

- **Actors involved and beneficiaries**

Cruises in the Atlantic Islands is an association formed by the Port Authorities of Madeira, Cape Verde, Las Palmas and Santa Cruz de Tenerife to promote cruises in the ports of Funchal (Madeira), Arrecife (Lanzarote), Puerto del Rosario (Fuerteventura), Las Palmas (Gran Canaria), Mindelo (Sao Vicente), Praia (Santiago), Santa Cruz de La Palma (La Palma), San Sebastián de La Gomera (La Gomera), Los Cristianos (Tenerife), La Estaca (El Hierro) and Santa Cruz de Tenerife (Tenerife).

- **Possible contribution to the fulfilment of BG objectives**

This is a private initiative with great potential to trigger further development of the activity in the four archipelagos.

- **Website**

http://www.cruisesintheatlanticislands.com/

7 Mapping main stakeholders

Table 5 — Main stakeholders surveyed in Macaronesia

OR/sea basin ([11])	Maritime sector(s)	Name of the stakeholders	Main roles and responsibilities	Address	Website
Macaronesia	All maritime sectors	Maritime Cluster of the Macaronesia	To promote growth and sustainable employment, according to the maritime strategy for the Atlantic.	C/ León y Castillo, 89 35004—Palmas de Gran Canaria Las Palmas España	http://plocan.eu/b3m/b3m-n8.pdf
Canary Islands	All maritime sectors	PROEXCA	To improve competitiveness of the Canarian enterprises trough support to their internationalization and to attract investment to the archipelago.	C/Emilio Castelar 4, 5ª planta. 35007 — Las Palmas de Gran Canaria España	http://www.proexca.es/
Canary Islands	R & D	Instituto Tecnológico de Canarias (ITC)	The Technological Institute of the Canary Islands is a public enterprise, and attached to the Department of Employment, Industry and Trade of the regional government. ITC conducts Research, Development and Innovation at the regional level. ITC supports the Island's integral development through the implementation of practices and the deployment of R&D projects in renewable energies, desalination of sea water, biotechnology, amongst others.	Calle Cebrian 3, 35003 Las Palmas Gran Canaria	www.itccanarias.org
Canary Islands	Maritime sectors : ship repair, aquaculture, fishing, nautical sports, recreation, tourism	Maritime Cluster	The cluster aims to promote the development and international competitiveness of the Canarian maritime sector. It integrates, strengthens, and ensures the sustainability of the business and agents of the value chain in the maritime sector. It promotes their international presence and increases its technological and innovative level, in line with development policies and social demands. The sectors under the umbrella of the cluster include marine and/or maritime industry, in consistence with the EU's 'Blue Growth'. The sectors covered by the cluster include ship repair, infrastructures and maritime transport, aquaculture, marine biotechnology, extractive fishing, nautical sports and recreation, R&D in renewable offshore energies, and the auxiliary maritime services.	Calle León y Castillo, 89 — 4ª planta, 35004 Las Palmas de Gran Canaria	www.clustermc.es
Canary Islands	R&D	PLOCAN (Plataforma Oceanográfica de Canarias)	The Oceanic Platform of the Canary Islands is a Unique Scientific and Technical Infrastructure (ICTS) dedicated to science and technology in the marine and maritime sector in the region. Its mission is the cost-effective combination of services such as observatories, test beds, underwater vehicles support, information technology, and	Carretera de Taliarte, s/n, 35214 Telde, Las Palmas	www.plocan.eu

([11]) Stakeholders with relevance to promoting blue growth in the entire Macaronesia.

OR/sea basin ([11])	Maritime sector(s)	Name of the stakeholders	Main roles and responsibilities	Address	Website
			training and innovation hub. Its activity is focused on renewable energies, underwater robotics, safety and security, ocean observation, environmental impact and pollution control or technologies associated with harnessing marine resources.		
Canary Islands	Education, R&D	Universidad de las Palmas de Gran Canaria (ULPGC)	The University of the Las Palmas de Gran Canaria covers a large array of fields of knowledge. It also runs the Scientific and Technological Park. The ULPGC's research fields include marine science, computer, telecommunications and electronic technology; economics; development cooperation; renewable energies; environmental conservation and tourism, among others.		www.ulpgc.es
Cape Verde	All maritime sectors	SIA (Environmental Information System)	Their task is planning and managing the development, protect optimal and integrated use of natural resources and their relationship with the national environmental policy, coordinate cross-cutting systems with impact on ecological sustainability and protection of natural biodiversity of the country, environmental education, ensuring the participation and information to the public, non-governmental organizations and companies	Direcção Geral do Ambiente CP 115—Fazenda, Praia Cabo Verde	http://www.sia.cv/
Cape Verde	All maritime sectors	Maritime Cluster of Cabo Verde	Consolidation of the Cape Verde position in the Mid-Atlantic lies in strengthening its maritime activities; the country has set a Strategic Agenda for the Maritime Cluster. It has therefore set up a small and dedicated organization to promote the Maritime Cluster in the broadest sense, NOCMAR	Not available.	Not available
Cape Verde	All maritime sectors	University of Cape Verde	The University of Cape Verde is a public institution of higher education located in Praia, Cape Verde. The university was founded in 2006 through the merger of two colleges that operated in different locations: the Institute of Education (ISE), located in Playa, and the Institute of Marine Sciences and Engineering (ISECMAR) located in Mindelo. In 2007, a third school officially joined to the university: the National Institute for Agricultural Research and Development (INIDA), located in São Jorge dos Órgãos.	Av. Santo Antao, Praia, Cabo Verde	www.unicv.edu.cv
Autonomous Region of Madeira	Marine research	Universidade da Madeira Centro de estudos da Macaronésia	Research on biotechnology and biological resources	Campus da Penteada 9020 Funchal	www.uma.es
Autonomous Region of Madeira	Research& development	Observatório Oceânico da Madeira	Research and monitoring of the ocean	Edifício Madeira Tecnopolo, piso 0 9020 -105 Funchal	http://oom.arditi.pt/
Autonomous Region of	Research& developme	Agência Regional para	Private no-profit association that aims to strengthen collaboration and links between its members the scientific and business	Edif. Madeira Tecnopolo, Piso 2	http://www.arditi.pt/

OR/sea basin ([11])	Maritime sector(s)	Name of the stakeholders	Main roles and responsibilities	Address	Website
Madeira	nt	o Desenvolvi-mento da Investigação Tecnologia e Inovação de Madeira (ARDITI)	community and to promote support research and development	Caminho da Penteada 9020-105 Funchal	
Autonomous Region of Madeira	All	ACIF- Câmara do Comércio e Industria da Madeira	Chamber of commerce and industry	Rua dos Aranhas, 24-26 9000-044 FUNCHAL	http://www.acif-ccim.pt/
Autonomous Region of Azores	Marine research	Centro do Clima, Metorologia e Mudanças globais, Universidades dos Açores	Research	Rua Capitão João d'Ávlia—Pico da Urze 9700-042 Angra do Heroísmo Açores, Portuga	http://www.dca.uac.pt
Autonomous Region of Azores	Marine research	IMAR / Universidade dos Açores	Research	Rua Prof. Dr. Frederico Machado — 9901-862 Horta — Portugal	http://www.imar.pt/

Mapping Blue Growth and assessing the potential

8 Growth drivers and barriers

Growth drivers and barriers are analysed for existing maritime sectors and maritime sectors with potentials in terms of R&D, Access to finance, Infrastructure, Education, training and skills, Public engagement and Local cooperation ([12]).

	Fisheries and aquaculture	
	Drivers to growth	**Barriers for growth**
R&D	Institutes conducting sound R&D, especially in the EU' ORs.	N/A
Access to finance	EU finance for productive activities at OR level; EU funding for transnational cooperation in the whole sea basin	N/A
Infrastructure	The OR possesses R+D capabilities and training infrastructures. Infrastructures for fisheries seem sufficient for the current state of the activity at least for the ORs. Aquaculture and especially algae production require infrastructure to add value to the production. Investments in infrastructure in Cape Verde are the backbone of its export oriented processing industry.	The fleet is aged and this may require investments for catching news species. This may also have implications for preservation and security on board. The OR markets have preference for fresh fish and this may impede development of the processing industry.
Education, training and skills	Education at upper and medium level is available, especially in the ORs.	Low qualification and lack of generational relief are reported.
Public engagement	National strategies for fisheries and aquaculture are being carried out in the ORs.	Administrative processes to obtain aquaculture permissions Strong environmental considerations retard introduction of new species Quotas and conservation and management measures
Local cooperation	Cooperation at the level of the entire sea basin is being conducted to develop new fisheries e.g. deep-sea fishing. It is worth commenting that the OR fishing sectors are active in the SWW AC. The sectors are also discussing their involvement in the potential Outermost Region AC.	N/A

	Desalination of seawater	
	Drivers to growth	**Barriers for growth**
R&D	Growth of tourism in Canarias and Cape Verde. EU funding for transnational cooperation is available for the entire sea basin. Canarias's institutions conduct R&D and innovation and constitute an asset in the sea basin for the development of this sector in other regions e.g. Cape Verde.	N/A
Access to finance	Finance in available for the Canarias sector.	N/A
Infrastructure	Large infrastructure in Canarias.	Taxes on self-consumption of renewable energies
Education, training and skills	Skills are available and considerable knowledge has been gathered in the	Lack of specialised education offer

([12]) i.e. synergies and conflicts at local level with other activities, be they maritime or non-maritime.

	Desalination of seawater	
	Drivers to growth	**Barriers for growth**
	Canarias sector.	
Public engagement	In Canarias, water plans are developed at the level of the insular governments.	N/A
Local cooperation	Cooperation at the level of the entire sea basin is being conducted to develop this sector in Cape Verde	N/A

	Maritime transport	
	Drivers to growth	**Barriers for growth**
R&D	EU funds for transnational cooperation	N/A
Access to finance	Funds for transport improvements are available from ERDF	N/A
Infrastructure	Infrastructures have been improved with EU support.	Transport of passenger to small islands need to be reinforced.
Education, training and skills	Training offer in Canarias and Azores. Regional human capacity specialised in freight shipping, logistics and port operations.	Training offer in Madeira and Cape Verde
Public engagement	N/A	N/A
Local cooperation	Cooperation at the level of the entire sea basin is being conducted in the Macaronesian Maritime Cluster, where private and public actors of the three archipelagos participate.	N/A

	Ports	
	Drivers to growth	**Barriers for growth**
R&D	EU funds for transnational cooperation	N/A
Access to finance	Funds for port improvements are available from ERDF	N/A
Infrastructure	Investment in ORs ports have been made with EU support	Moorings and other infrastructures for growing demands of nautical activities.
Education, training and skills	Training offer in Canarias and Azores	Training offer in Madeira and Cape Verde
Public engagement	Port specialization has been conducted in the OR ports	N/A
Local cooperation	Cooperation at the level of the entire sea basin is being conducted in the Macaronesian Maritime Cluster, where private and public actors of the three archipelagos participate.	Lack of space in some ports can give raise to conflicts amongst diverse uses of the port space.

	Ship building and repair	
	Drivers to growth	**Barriers for growth**
R&D	ORs universities and institutes conduct R&D, especially in Canarias.	N/A
Access to finance	Finance is available in Canarias.	Investments for training infrastructures are costly.
Infrastructure	Canarias ports provide facilities for the ship repair activity.	N/A
Education, training and skills	Training offer from universities and federation of the metal sector.	In spite of the efforts conducted technical means and especially human capacities are still limited to address current needs of oil platforms e.g. technical skills, English command.
Public engagement	Increasing awareness in the ORs about the potentialities of the ship repair sector.	N/A

	Ship building and repair	
	Drivers to growth	**Barriers for growth**
Local cooperation	This is high at the level of Canarias where the federations of metal entrepreneurs are taking action to improve the training offer.	N/A

	Coastal tourism	
	Drivers to growth	**Barriers for growth**
R&D	EU funds for transnational cooperation are in place for the whole sea basin. R&D in place for the ORs. Initiatives are being conducted in developing new routes for nautical activities in Cape Verde.	N/A
Access to finance	At OR level.	Highly indebted sectors.
Infrastructure	A large offer of hotels and other facilities e.g. marinas, golf yards, especially in Canarias.	Obsolete facilities in the oldest touristic sites in Canarias and Madeira.
Education, training and skills	Available, especially in the EU's ORs.	The activity is highly seasonal and consequently its labour needs. Human resources have empirical knowledge but in many cases lack professional qualification.
Public engagement	Transnational cooperation promoted by the UE. Due to the strategic role in the archipelagos economies the sector is highly relevant in the agenda of politicians e.g. RIS3 in Canarias.	N/A
Local cooperation	Cooperation at the level of the sea basin in the framework of the Maritime Cluster of the Macaronesia and in the project Technological Platform, especially with regards to nautical tourism.	N/A

	Cruise tourism	
	Drivers to growth	**Barriers for growth**
R&D	N/A	N/A
Access to finance	N/A	N/A
Infrastructure	Madeira and Canarias have made important efforts to improve infrastructures and the offer of packages to attract cruise tourists.	Lack of facilities for cruise tourists doing a stop in the islands, especially in relation to urban services and touristic offer.
Education, training and skills	Training offer in Canarias and Azores.	Lack of training offer in Madeira.
Public engagement		
Local cooperation	Ongoing cooperation at the level of the Macaronesia. Space at ports especially in Portuguese ORs.	N/A

8.1 Synoptic view of legislative, non-legislative and financial growth drivers and barriers

The following table reports a synoptic view of main growth drivers and barriers grouping them under legislative, non-legislative and financial frames.

Growth drivers			
ACTIVITY	LEGISLATIVE	NON-LEGISLATIVE	FINANCIAL
Fisheries and aquaculture	Participation of the OR fishing sectors in the SWW AC provided the opportunity to promote their interests before the EU.	• Institutes conducting sound R&D, especially in the EU' ORs. • The OR possesses R+D capabilities and training infrastructures. • Education at upper and medium level is available, especially in the ORs. • National strategies for fisheries and aquaculture are being carried out in the ORs. • Cooperation at the level of the entire sea basin is being conducted to develop new fisheries e.g. deep-sea fishing.	• EU finance for productive activities at OR level is a key growth driver. • ERDF funding of transnational cooperation initiatives in R&D is of key importance for the sea basin. • FDI in infrastructure in Cape Verde is the backbone of its export oriented processing industry.
Desalination of seawater		The thorough expertise of the Canary Islands is being transferred to Cape Verde trough participation in EU funded cooperation projects.	
Maritime transport		• Training offer in Canarias and Azores in freight shipping, logistics and port operations. • Cooperation at the level of the entire sea basin is being conducted in the Macaronesian Maritime Cluster, where private and public actors of the three archipelagos participate.	In the ORs, funds for transport improvements are available from ERDF.
Ports		• Training offer in Canarias and Azores in freight shipping, logistics and port operations. • Cooperation at the level of the entire sea basin is being conducted in the Macaronesian Maritime Cluster, where private and public actors of the three archipelagos participate.	Funds for OR's ports improvements are available from ERDF.
Ship building and repair		• ORs universities, institutes and federation of entrepreneurs offer R&D, education and technical training, especially in the Canary Islands. • The Canary Islands ports provide ship repair infrastructures of key interest for the sea basin, supporting the growth of the shipping, cruise, and nautical activities. They are also of potential interest for possible blue energy activities.	
Coastal tourism		• R&D in place for the ORs. • Initiatives are being conducted to developing new routes for nautical activities in Cape Verde.	EU funds for transnational cooperation are in place for the whole sea basin (ERDF's MAC program).
Cruise tourism		• Madeira and Canarias have made important efforts to improve infrastructures and the offer of packages to attract cruise tourists. • Training offer in Canarias and Azores. • Ongoing cooperation at the level of the Macaronesia.	

Barriers to growth			
ACTIVITY	LEGISLATIVE	NON-LEGISLATIVE	FINANCIAL
Fisheries and aquaculture	• In the ORs, heavy administrative burdens for the aquaculture activity are in place (licensing and concession of spaces).	• The fleet is aged. This may have implications for fish preservation and crews' security on board. • The OR markets have preference for fresh fish and this may impose a barrier for the development of the processing industry. • Low qualification and lack of generational relief are reported, especially in the ORs. • Long administrative processes for obtaining concessions are regarded as a strong limitation in the ORs. • Difficulty to access to veterinary medicines for the farmed species, which limit the capacity of tackling possible fish diseases in due time.	• There is a lack of credit by private financial intermediaries. • Lack of credit to renew fishing vessels.
Desalination of seawater		Investments are high and not affordable for Cape Verde without in kind transferences and financial support of developed nations.	
Maritime transport		• Lack of recreational relief is reported in the ORs. • Transport of passenger to small islands need to be reinforced with more frequencies. • Training offer in Madeira and Cape Verde is regarded as limited, especially in relation to operational functions and shipping and logistics management.	
Ports		• Training offer for port personnel is limited in Madeira and Cape Verde. • Lack of space in some ports can give raise to conflicts amongst diverse users of the port space. • Moorings and other infrastructures are required for growing demand of nautical activities.	
Ship building and repair		• Investment in infrastructures and training is costly. • Technical means, and especially human capacities, are still limited to address current needs of offshore activities in the ORs e.g. technical skills, English command.	
Coastal tourism		• Obsolete facilities in the oldest touristic sites in Canarias and Madeira. • The activity is highly seasonal and consequently its labour needs. • Human resources have empirical abilities but in many cases lack formal qualifications.	
Cruise tourism		• Seasonality is regarded as a strong limitation for the activity in the sea basin, especially for the Canary Islands. • Limited facilities for cruise stop overs, especially in relation to urban services and touristic offer.	

9 Future potential for mature and pre-developing activities

9.1 Fisheries and aquaculture

- **Potential for employment**

In the Macaronesia, employments are offered basically in the harvesting sector where low and medium level qualified workers are employed. Regarding the processing sector, Cape Verde is likely the most developed in the whole sea basin due to the foreign investments and the relevance in the national economy (70 % - 80 % of exports). That sector offers potential for employment for low and medium qualification workers. In general, aquaculture in the sea basin offers job opportunities for medium and high qualified personnel. There is also a potential to diversify the fishing activity thus reducing pressure over the resources e.g. pesca-tourism. The latter and other forms of tourism where fishing skippers and crews could participate require medium qualification including security on board, observation of marine mammals and birds, and cultural topics. Pesca-tourism requires stretching bonds with touristic operators.

- **Potential for value added**

The sea basin has a potential for new fisheries e.g. deep-water species. Other species of low value e.g. some small pelagics could sustain a fishmeal industry supplying feed to the aquaculture activity. The fishing fleet is in general aged; a competitive fishing sector requires renovation to increase value adding. It is difficult to assess growth in employment in the harvesting sector due to the limitations imposed by quota restrictions and fleet restructuration, particularly in the case of the EU' ORs. Growth in the processing sector seems more likely in the Cape Verde case which is the most industrialized sector and has attracted foreign investment. Expansion of aquaculture and especially the production of algae provide opportunities for a biotechnology industry. This sector seems more likely to growth in the EU's OR due to the presence of sound R+D activities although high investments are regarded as a constraint. The sea basin market is particularly interesting for the commercialization of fresh and frozen fish. Considering that most of the fish production of Cape Verde is processed and exported to markets in developed regions a lack of fish supply may emerge in that archipelago. Exploitation of new species, especially deep-water species, to satisfy potential demand in Cape Verde and in the ORs also arises as an alternative for protein supply.

- **Potential for competitiveness**

Availability of human resources adequately skilled.	General low qualification in the harvesting sector. Although a training offer exist, especially in the Canary Islands. Generational relief seems a problem for the sector in the whole sea basin, especially in the EU's OR.
Potential for attracting FDI.	Some of the sectors such as the algae production may attract FDI. In Cape Verde the processing sector currently attracts FDI.
Availability of public funds.	EMFF available for the EU' ORs. The MAC program is available for R+D for the entire sea basin.
Availability of support sectors / facilities (e.g. research, infrastructures, etc.).	EU's OR possesses R+D capabilities and training infrastructures. Infrastructures for fisheries seem sufficient at least for the EU's OR. Aquaculture and especially algae production require infrastructure to add value to the production.
Potential in terms of positioning in the international market.	Algae production and aquaculture in general have the potential to position in the international market.

9.2 Desalination

- **Potential for employment**

Currently, this activity is only conducted in Madeira and the Canary Islands. It employs medium and high qualified personnel. Development of the sector in Cape Verde due to scarcity of natural water will demand qualified personnel. There is a potential for cooperation in the region in qualification of personnel. Training offer for this activity is limited even in the Canary Islands, which is one of the leading regions in the world in desalination of seawater.

- **Potential for value added**

The activity has a potential for growth in the sea basin but especially in the Canary Islands and Cape Verde. The low availability of natural water together with the increasing needs of the population and tourism require expansion of the sector in the Canary Islands and development of the sector in Cape Verde. The expertise gathered in the Canary Islands is currently being exported in the form of technical assistance to third countries and constitute an asset in the sea basin, being highly useful to develop the activity in Cape Verde.

- **Potential for competitiveness**

Availability of human resources adequately skilled.	Human resources are available in Madeira and the Canary Islands. Due to the experience accumulated in five decades of desalination of seawater, large sector and R+D facilities Canarias concentrates most of the expertise in the sea basin. This offers a basis for technology transference and capacity building.
Potential for attracting FDI.	Little room for FDI in Canarias and Madeira, where the activity has traditionally been in hands of regional private capitals and governments. Cape Verde might have some potential to attract FDI. This need to be assessed.
Availability of public funds.	MAC programme available for the entire sea basin. Funds for infrastructure are available at regional level.
Availability of support sectors / facilities (e.g. research, infrastructures, etc.).	The increasing needs require expansion in Canarias and development in Cape Verde. R+D capabilities are in place especially in Canarias.
Potential in terms of positioning in the international market	The expertise gathered in the Canary Islands is a referent in the world.

9.3 Maritime transport

- **Potential for employment**

The growing demand of maritime transport requires training of human resources to be reinforced/develop, especially in Azores and Cape Verde. Support services linked to ships, freight and crews, and more frequencies to small islands are required. The growing demands of nautical activities make evident the lack of infrastructures and of qualified personnel to attend the needs of this emerging activity.

- **Potential for value added**

Improvement of energy efficiency in freight transport and creation of synergies between the ferry and tourism sector might have implications on value adding in the sea basin.

- **Potential for competitiveness**

The strategic position of the four archipelagos in the Atlantic Ocean confers advantages for the development of freight maritime transport. Azores and especially Canary Islands have a great potential for becoming logistic hubs in the Mid-Atlantic.

Availability of human resources adequately skilled.	The growing demand of maritime transport and ports services requires training of human resources to be reinforced/develop, especially in Azores and Cape Verde.
Potential for attracting FDI.	N/A.
Availability of public funds.	Public funds are available for cooperation.
Availability of support sectors / facilities (e.g. research, infrastructures, etc.).	Large investments have been made to turn OR ports into competitive infrastructures able to take advantage of the strategic location of the archipelagos. Cooperation between the ports of the region will be required to establish a network of ports in the sea-basin.
Potential in terms of positioning in the international market	Being in the crossroads of maritime routes and having a privileged position in relation to three continents the Macaronesian archipelagos have a great potential to position in international markets.

9.4 Ports

- **Potential for employment**

The privileged position of the archipelagos, the growth of passengers and goods transport and of naval and nautical activities will have positive implications for employment creation in the ports of the sea basin. The construction of new port areas (marina, fishing and commercial facilities) and port services would create direct and indirect jobs.

- **Potential for value added**

There is a potential to increase value added in the transport of goods, naval sector, nautical activities, and bunkering, amongst others.

- **Potential for competitiveness**

Availability of human resources adequately skilled.	The growing demand of ports services requires training of human resources to be reinforced/develop, especially in Azores and Cape Verde.
Potential for attracting FDI.	FDI have been made in the main ports of Canarias, e.g. ship yards for yachts.
Availability of public funds.	Public funds are available for cooperation.
Availability of support sectors / facilities (e.g. research, infrastructures, etc.).	ORs ports infrastructures have been modernised. Ports specialisation has been carried out. It would be worth introducing infrastructures for alternative fuels in ships (LNG).
Potential in terms of positioning in the international market	The OR's ports are already well positioned at international level. There is still room, however, to turn the ports into centres for transport of goods, cruises, nautical activities, fishing and naval repair, humanitarian aid, and other services.

9.5 Ship building and repair

- **Potential for employment**

Shipbuilding is only relevant for Azores and the Canary Islands although the activity is only marginal. Regarding ship repair high and medium skill jobs are being created, especially in connection to the offshore activities. The growth of the activity requires qualifications and professionalization. It is worth pointing out, however, that the activity of offshore platform repair is subject to the fluctuations of oil price (see the Canary Islands report).

- **Potential for value added**

This activity of ship repair has a great potential for value adding in the Canary Islands. It can generate a critical mass that can sustain future developments in marine technology such as, for example, blue renewable energies (offshore wind and wave energy). These constitute an asset in the sea basin to explore the development of marine renewable energies sector. Repair of recreational boats, including yachts, has good perspectives in the sea basin provided that the nautical sector is growing.

- **Potential for competitiveness**:

Availability of human resources adequately skilled.	Only relevant at the level of the EU' ORs (see the respective reports).
Potential for attracting FDI.	This is particularly relevant for Canarias.
Availability of public funds.	ERDF funds for port infrastructure have positive implications for this activity but there are not direct public investments.
Availability of support sectors / facilities (e.g. research, infrastructures, etc.).	These are available especially in Canarias e.g. universities, metal federations. Training infrastructures demand further investments. Currently, foreign capitals are doing investments in training facilities in Canarias.
Potential in terms of positioning in the international market	The EU' ORs, especially Canarias and to a lesser degree Azores has a high potential for positioning in international markets.

9.6 Coastal and cruise tourism

- **Potential for employment**

Qualification of the human resources and training are required. Training activities are being conducted to respond to punctual demands but a steady offer of training for low and medium level positions seems lacking. The seasonal pattern of the activity may have had implications in the lack of qualification and professionalization in the activity. New forms of tourism such as pesca-tourism, coastal culture, mammals and bird sighting require training and certification of operators.

- **Potential for value added**

Value added could be increased by diversification of the touristic offer. The old model of sun & beach is currently being replaced by a more extensive use of the marine space and of its resources.

- **Potential for competitiveness**:

Availability of human resources adequately skilled.	There is a reported lack of qualified human resources in the whole sea basin. There is, however, an educational offer even at graduate level in the EU's ORs. Strategic tourism plans e.g. of the Canary Islands point out the need for a steady supply of training activities.
Potential for attracting FDI.	These activities have traditionally attracted FDI. The Cape Verde tourism has the largest potential since this activity is still developing.
Availability of public funds.	MAC program for R+D and innovation in the whole sea basin.
Availability of support sectors / facilities (e.g. research, infrastructures, etc.).	Large infrastructures in the EU's OR although obsolescence in accommodation facilities are reported, especially in Madeira and Canarias. Investments have been made in improving cruise infrastructures e.g. in the Canarias and Madeira. There is still a need to provide adequate berthing for cruises e.g. in Azores. There seems to be a need to improve infrastructures and services for cruise stopovers in the ORs.
Potential in terms of positioning in the international market	The EU's ORs are well positioned in international markets. New forms of tourism which use the entire marine environment as a source of leisure experiences have the potential to increase the position of these archipelagos. In turn, Cape Verde has a great potential to develop further in particular in relation to nautical tourism. The tourism offer associated to cruisers has also a potential and cooperation in this matter will make the sea basin more competitive.

9.7 Blue energies (pre-development)

- **Potential for employment**

The activity is conducted in Azores and the Canary Islands. It will demand medium and highly qualified personnel for the operation of the energy devices. The R+D activity will require graduate and post-graduate personnel.

- **Potential for value added**

This activity has a great economic potential in the Canary Islands and Azores as one of the forms to reduce dependency on fossil energies.

- **Potential for competitiveness**

Availability of human resources adequately skilled.	Highly qualified engineers and scientists are available in EU's ORs, especially in the Canary Islands.
Potential for attracting FDI.	Not assessable at this stage.
Availability of public funds.	MAC programme for R+D for the entire sea basin.

Availability of support sectors / facilities (e.g. research, infrastructures, etc.).	These are available only in Azores and Canarias.
Potential in terms of positioning in the international market	The technologies being developed in the Canary Islands have a potential for positioning in international markets. This archipelago is a natural lab for the testing of related technologies.

9.8 Blue biotechnologies (pre-development)

- **Potential for employment**

This activity seems more viable in the Canary Islands and Madeira. It is highly intensive in technology and capital so potential employment opportunities would likely include university graduates and postgraduates and lab technicians.

- **Potential for value added**

Currently, the activity of microalgae production has little value adding since products are commercialized in powder and dehydrated. There is a high potential for value adding trough extraction of bioactive compounds.

- **Potential for competitiveness**

Availability of human resources adequately skilled.	Institutes in the EU's ORs conduct sound research and development on algae identification and production.
Potential for attracting FDI.	High costs of investments and transport may impose a barrier for foreign investments.
Availability of public funds.	In the EU's ORs the sector of algae production is still considered as an aquaculture sector thus eligible for funding under the EMFF.
Availability of support sectors / facilities (e.g. research, infrastructures, etc.).	Microalgae production in Canarias and Madeira require use of brine and provision of CO2. Expansion of this production to encompass a large biotechnology industry will require these infrastructures plus equipment to obtain bioactive compounds.
Potential in terms of positioning in the international market	Bioactive compounds have an increasing global demand for food, cosmetic and pharmacy industries. USA (Hawaii) and China are two of the world leaders in this sector. Finding a place in such competitive market is costly but natural conditions and qualified personnel constitute key inputs that are already in place.

9.9 Synoptic view of the future potential by activity

The following table reports a synoptic view of main future potential for each activity grouping them by 'Potential for employment', 'Potential for value added' and 'Potential for competitiveness'.

ACTIVITY	POTENTIAL FOR EMPLOYMENT	POTENTIAL FOR VALUE ADDED	POTENTIAL FOR COMPETITIVENESS
Fisheries and aquaculture	• The fishing and processing sectors offers potential for employment for low and medium qualification workers. • In turn, aquaculture in the sea basin offers job opportunities for medium and high qualified personnel. • There is also a potential to diversify the fishing activity thus reducing pressure over the resources e.g. pesca-tourism. This activity will require a workforce with medium qualification including security on board, observation of marine mammals and birds, and cultural topics. • Growth of aquaculture may contribute to employment positive spill over effects on conservation, transport, packing and distribution.	• This potential seems greater in the Cape Verde case, which is the most industrialized sector and attracts FDI. • Expansion of algae production provides opportunities for a biotechnology industry. This sector seems more likely to growth in the EU's OR due to the presence of sound R+D activities although high investments are perceived as a constraint. • New species of high value such as corvina have a potential to add value to the activity e.g. in Canarias. • Potential resources such as deep-water resources for Canarias, shrimps and prawns for Madeira and other resources with high and low value have the potential to add value to the sector for fresh and processed human consumption and also for animal feed.	• Broadly speaking, low qualification levels are reported in the harvesting sector. • Generational relief seems a problem for the sector in the whole sea basin, especially in the EU's OR. • Sectors such as the algae production may attract FDI. • In Cape Verde the processing sector currently attracts FDI. • The MAC program is available for R+D for the entire sea basin. • EU's OR possesses R+D capabilities and training infrastructures. • Due to the natural conditions of the sea basin finfish reach commercial size before than in other sea basins. • Algae production and aquaculture in general, have the potential to position in the international market.
Desalination	• It employs medium and high qualified personnel. • Development of the sector in Cape Verde due to scarcity of natural water will demand qualified personnel. • Training offer for this activity is limited even in the Canary Islands.	• The activity has a potential for growth in the sea basin, especially in the Canary Islands and Cape Verde, which offers a basis for technology transference and capacity building. • The expertise gathered in the Canary Islands is currently being exported in the form of technical assistance to third countries and constitute an asset in the sea basin, being highly useful to develop the activity in Cape Verde.	• Human resources are available in Madeira and the Canary Islands. • Little room for FDI in Canarias and Madeira, where the activity has traditionally been in hands of regional private capitals and governments. • MAC programme available for cooperation and R&D in the entire sea basin. • R+D capabilities are in place, especially in Canarias.
Maritime transport	• The growing demand of maritime transport requires specialised workers in Madeira e.g. shipping and logistics management and in Cape Verde.	Improvement of energy efficiency in freight transport and creation of synergies between the ferry and tourism sector might have implications on value adding in the sea basin.	• Although qualified personnel is available in the Canary Islands and Azores the growing maritime transport requires training of human resources to be

ACTIVITY	POTENTIAL FOR EMPLOYMENT	POTENTIAL FOR VALUE ADDED	POTENTIAL FOR COMPETITIVENESS
	• Support services linked to ships, freight and crews, and more frequencies to small islands are required. • The growing demand of nautical activities makes evident the lack of infrastructures and of qualified personnel to attend the needs of this emerging activity.		reinforced/develop, especially in Madeira and Cape Verde. • Public funds are available for cooperation at the sea basin level (MAC program). • Cooperation between the ports of the region will be required to establish a network of ports in the sea-basin. • Being in the crossroads of maritime routes, and having a privileged position in relation to three continents, the Macaronesian archipelagos have a great potential to position in international markets.
Ports	• The privileged position of the archipelagos, the growth of passengers and goods transport and of the naval and nautical activities will have positive implications for employment creation in the ports of the sea basin. • The construction of new port areas (marina, fishing and commercial facilities) and port services would create direct and indirect jobs.	The growth of passenger and cargo maritime transport in the region has a potential to increase value added in the naval sector, nautical activities, and bunkering, amongst others.	• The growing demand of ports services requires training of human resources to be reinforced/developed, especially in Azores and Cape Verde. • FDIs have been attracted to the main ports of Canarias, e.g. ship yards for yachts. • Public funds are available for cooperation. • In the ORs ports infrastructures have been modernised. Ports specialisation has been carried out. • It would be interesting to introduce infrastructures for alternative fuels in ships (LNG). • The OR's ports are already well positioned at international level. There is still room, however, to turn the ports into hubs for transport of goods, cruises, nautical activities, fishing and naval repair, humanitarian aid, and other services.
Shipbuilding and repair	• High and medium skill jobs are being created for the ship repair activity. • The activity of offshore platform repair is subject to the fluctuations of oil price (see the Canary Islands report) and, thus, employment levels would tend to	• This ship repair in the Canary Islands can generate a critical mass that can sustain future developments in marine technology such as, for example, marine renewable energies (offshore and wave energy). This constitutes an asset in the sea basin to	• R&D and training are available especially in Canarias e.g. universities, metal federations. These are assets for the sea basin. Training infrastructures demand further investments. Currently FDI are doing investments in training facilities in

ACTIVITY	POTENTIAL FOR EMPLOYMENT	POTENTIAL FOR VALUE ADDED	POTENTIAL FOR COMPETITIVENESS
	fluctuate as well.	explore the development of marine renewable energies sector. • Repair of recreational boats, including yachts, has good perspectives in the sea basin provided that the nautical sector is growing.	Canarias. • The EU' ORs, especially Canarias and to a lesser degree Azores has a high potential for positioning in international markets.
Coastal and cruise tourism	• The growth of the activity in the entire sea basin may increase employment demand. However, there seems to be a lack of a steady offer of training for low-mid level positions. • New forms of tourism such as coastal culture, mammals and bird sighting, etc. may require personnel with appropriate qualifications.	Value added could be increased by diversification of the touristic offer. The old model of sun & beach is currently being replaced by a more comprehensive use of the marine space and of its resources.	• There is an educational offer even at graduate level in the EU's ORs. • These activities have traditionally attracted FDI. • The Cape Verde tourism has the largest potential since this activity is still developing. • The MAC program for R+D and innovation is available for the whole sea basin. • Obsolescence in accommodation facilities are reported in some ORs touristic sites. • Nautical activities have a high potential in the entire sea basin. There is also a high potential for touristic activities linked to nature and cultural experiences. • The tourism offer associated to cruisers has also a potential and cooperation in this matter will make the sea basin more competitive.
Blue energies	• The activity is conducted in Azores, and in Canarias at R&D level. It will demand medium and highly qualified personnel for the operation of the energy devices. • The R+D activity will require graduate and post-graduate personnel.	This activity has a great economic potential in the Canary Islands and Azores as one of the forms to reduce dependency on fossil energies.	• Highly qualified engineers and scientists are available in the ORs, especially in the Canary Islands. • The MAC programme for R+D is available for the entire sea basin. • The technologies being developed in the Canary Islands have a potential for positioning in international markets. This archipelago is a natural lab for the testing of related technologies.
Blue biotechnologies	This activity seems more viable in Canarias and Madeira. It is highly intensive in technology and capital so potential employment opportunities	The activity of microalgae production has little value adding. There is a high potential for value adding trough extraction of bioactive compounds.	• Institutes in the EU's ORs conduct sound research and development on algae identification and production. • High costs of investments and transport

ACTIVITY	POTENTIAL FOR EMPLOYMENT	POTENTIAL FOR VALUE ADDED	POTENTIAL FOR COMPETITIVENESS
	would likely include university graduates and postgraduates and lab technicians.		may impose a barrier for FDI. • Expansion of this production to encompass a large biotechnology industry will require heavy equipment to obtain bioactive compounds. • Bioactive compounds have an increasing global demand for food, cosmetic and pharmacy industries. This potential sector has advantages such as knowledge and natural conditions to become a player in this market.

10 Investments opportunities

Fisheries and aquaculture: In general terms, the fleet is aged so renovation of the fishing fleet would be required to enhance security on board and fish preservation facilities. Ports, especially those devoted to small scale fishing require improvement of the ice production capacity. Exploitation of deep-sea resources may arise as an investment opportunity i.e. new gear would be required. In Canarias, fish processing has little potential for growth. In contrast, the activity is already developed in Azores and Cape Verde and may still have room for development. The aquaculture activity requires production diversification, currently based on two species i.e. seabream and seabass. Cultivation of microalgae requires desalination of seawater facilities to obtain brine and water. In turn, water depuration facilities are also needed.

Desalination: Enlargement of desalination capacity will be needed in the near future due to growth of tourism and population, especially in the Canary Islands and Cape Verde. There is a need to reduce dependency on fossil energies for the production of water. The use of renewable energies to supply energy to the desalination plants requires investment in R&D and infrastructure.

Transport of passengers and ports: Interisland transport in the different archipelagos will require investment to satisfy the increasing demand of transport of passengers and goods. Port infrastructures will require improvement and expansion to satisfy the envisaged demand of services for transport of goods and passengers, nautical activities, ship repair, amongst others.

Shipbuilding and repair: The growth of yacht fit and repair businesses due to the emergence of the nautical activities will require enlargement of naval infrastructures, or reorganisation of port space, and investment in training of personnel to address the needs. Repair of offshore platforms and large vessels seems only viable in the Canary Islands. Training of personnel and international recognised qualifications are also needed.

Coastal and cruise tourism: Renewal of the accommodation infrastructure is required in the oldest touristic sites of the sea basin. It is envisaged that nautical activities will growth so increase of the offer of moorings in the entire sea basin should be carried out. Urban facilities and touristic offer for cruisers doing a call in the sea basin arise also as an opportunity for attracting cruises to the sea basin.

Blue biotechnology (pre-development stage): Extraction of the bioactive compounds, which are highly appreciated in international markets, could require substantial investments. This activity seems more viable in the ORs, especially in the Canary Islands.

Blue energies (pre-development stage): There is still a considerable effort to put marine wind farms and wave energy devices into commercial operation. In the ORs investments are specially required in the field of R&D.

11 Most relevant maritime activities at sea basin level

11.1 Fisheries and aquaculture

- **Workforce characteristics**

Workforce in most of the cases is national although in some cases (e.g. Canarian aquaculture) some specialists such as specialised divers are hired from other regions in Spain.

- **Administrative burden**

Fisheries activities are conducted in many cases outside the national EEZs of the countries concerned. These activities are regulated by the RFMOs. TACs are fixed by these organizations e.g. for blue fin tuna. Each EU OR and Cape Verde conducts their fishing activity on their own. There is not a joint exploitation of resources at the level of the sea basin.

- **Restrictive practices**

At the level of the entire sea basin the only restrictions imposed to fishing fleets are those established by the regulatory frameworks of the RFMOs.

- **Sources of finance**

The ERDF and specifically the MAC program offer funding for R+D and innovation in fisheries matters, which include the whole Macaronesia region. As an example of initiative the MARPROF-CV carried out a prospection of deep sea fishing resources in Cape Verde waters. In relation to infrastructures FEMP instruments are exclusively devoted to the EU's ORs.

11.2 Desalination

- **Workforce characteristics**

It is worth pointing out that three of the four archipelagos (Madeira, Canarias and Cape Verde) in the Macaronesia region have diverse degrees of dependency on desalinated seawater. The Canary Islands is the region with the largest needs due to water scarcity, large population and touristic sector. It possesses the largest desalination capacity and most comprehensive technical expertise. This expertise is been exported to other regions of the world, especially to Africa. It is envisaged, however, that increasing needs of desalination activities will require more qualified staff at the level of the Canary Islands and, consequently, for the whole sea basin. Cape Verde has also strong needs of water so technical expertise is required for the growth of this sector. Thus, cooperation between Canarias and Cape Verde seems necessary and in fact some experiences are in place e.g. the project ISLAGHUA. It is worth pointing that the Cape Verde is investing with support of third nations in developing the infrastructure needed for desalination of seawater.

- **Administrative burden**

There are no administrative burdens at the level of the entire sea basin. Administrative burdens are placed at the level of insular governments in the cases of the EU' OR.

- **Restrictive practices**

The sector is strongly regulated at local level. In the Canary Islands, desalination of seawater in combination with wind energy is facing imposition of taxes especially devised for this form of energy consumption. This is regarded as a barrier to growth provided that the mains constrain for production of water is the employment of fossil energy.

- **Sources of finance**

Sources for R+D are available from ERDF. Public financing for infrastructure is only available at the level of the EU's OR.

11.3 Maritime transport and ports

- **Workforce characteristics**

The transport and ports activities require qualified personnel. It is reported that the workforce is not sufficiently trained in Madeira and Cape Verde. In the case of the Canary Islands it seems that the workforce offer can currently satisfy the needs although it is envisaged that the expansion of activities will require more qualified personnel.

- **Administrative burden**

N/A.

- **Restrictive practices**

N/A.

- **Sources of finance**

Sources for R+D are available from ERDF. Public financing for infrastructure is only available at the level of the EU's OR.

11.4 Coastal and cruise tourism

- **Workforce characteristics**

Experienced workforce is available in Canarias and in the Portuguese ORs. Information about the level of expertise in Cape Verde is not available but seems limited due to the rather recent emergence of the touristic sector. It is reported that in general the touristic sector of the ORs require provision of training, especially for low and medium level positions. New forms of tourism such as cultural and nature tourism require trained personnel. Cruise tourism also requires experienced personnel, considering the expansion of the activity and the high profile of the cruise tourist that arrive in the diverse archipelagos.

- **Administrative burden**

N/A

- **Restrictive practices**

N/A

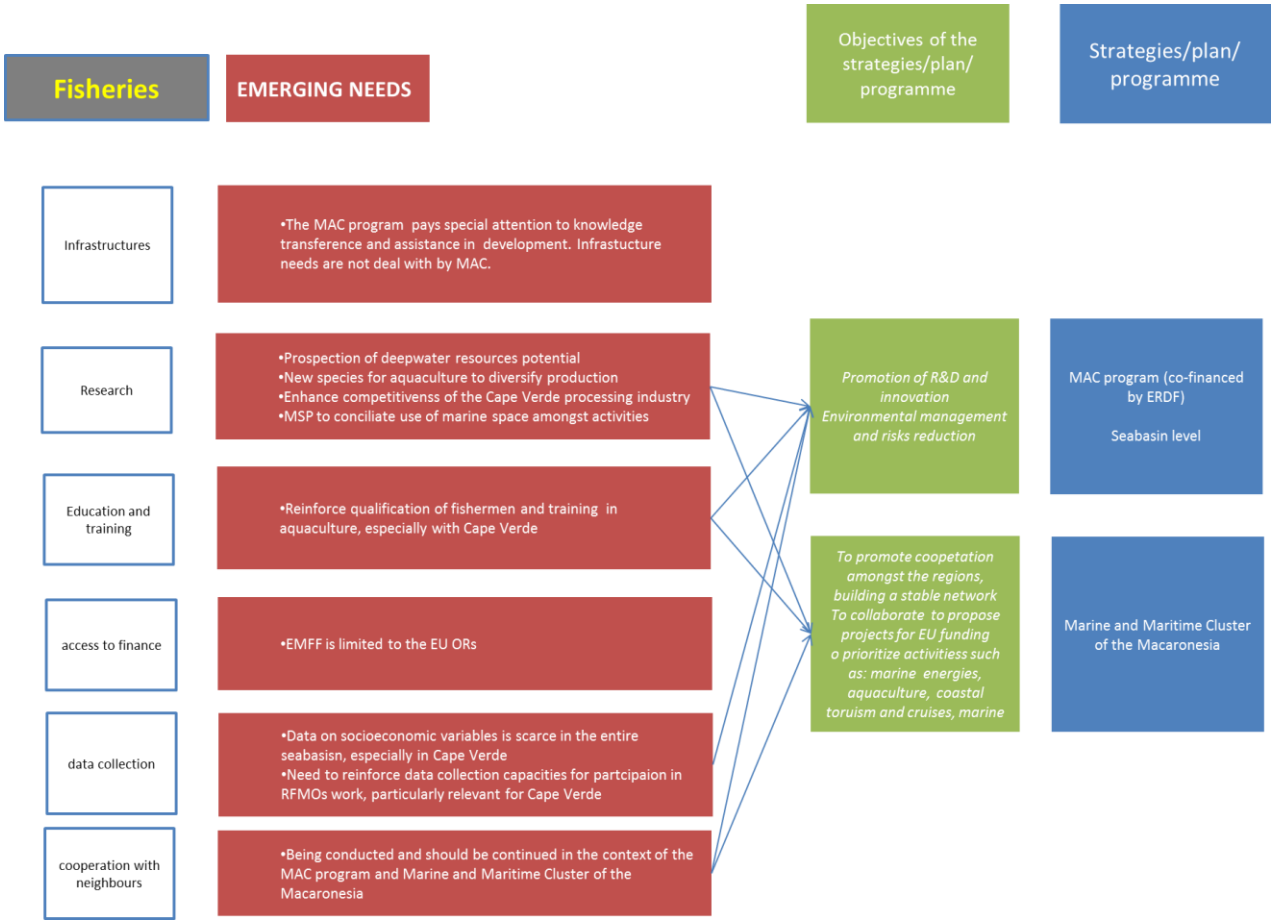
- **Sources of finance**

Sources for R+D are available from ERDF. Public financing for infrastructure is only available at the level of the EU's OR.

Identifying public and private intervention

12 Potential financial, legislative and non-legislative measures for stimulating growth

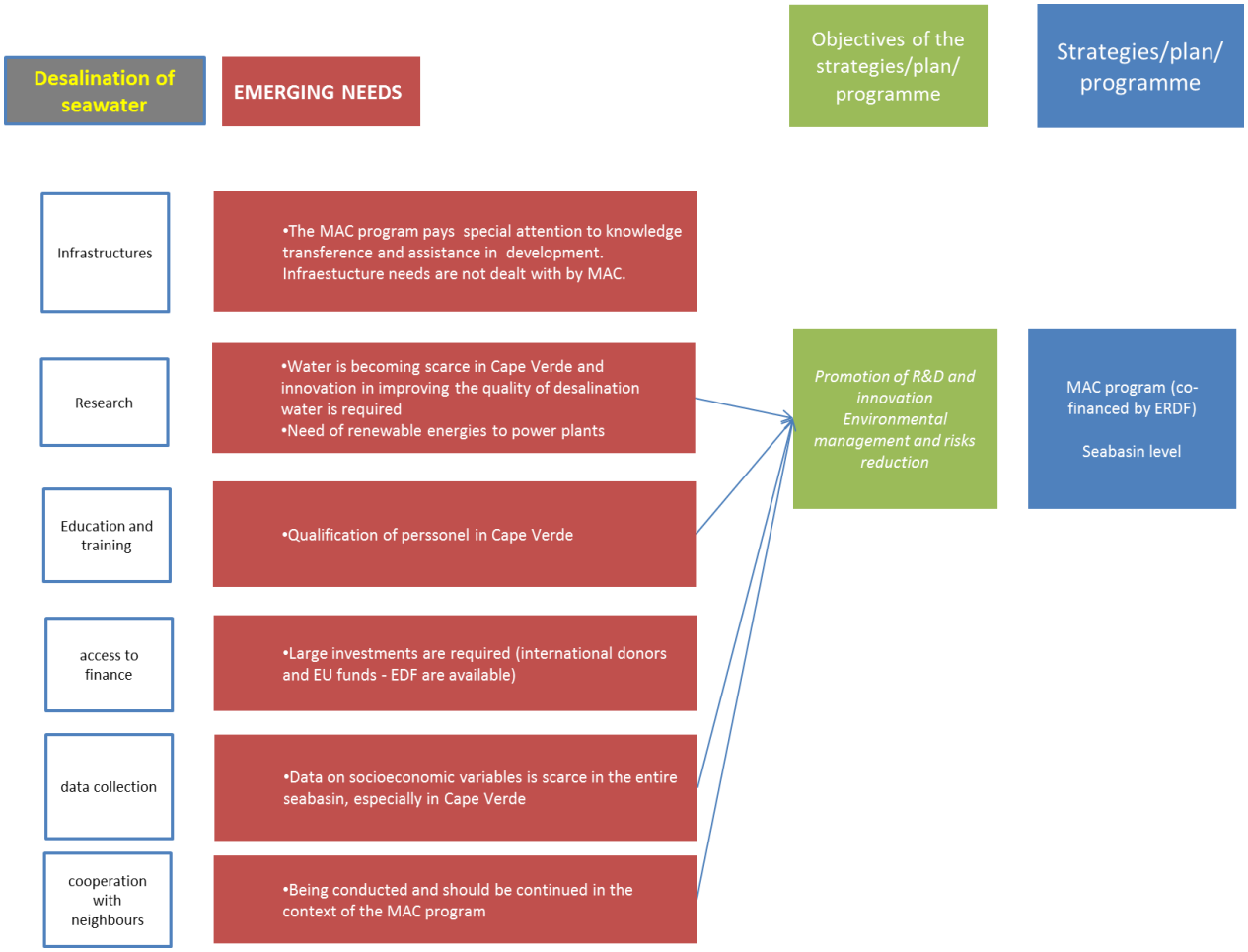
Figure 1 — Logical diagram for fishery



The ERDF's Macaronesian program (MAC) is of key importance in the sea basin to address the needs of the fishing sector and is especially relevant for Cape Verde, where technical assistance from the ORs takes place in the form of projects. It is worth pointing out that infrastructures are not funded by this program. Cape Verde has access to European Development Fund (EDF) for obtaining support for infrastructures. In relation to R&D and innovation, some of the needs in the sea basin are related to the identification of potential resources and **identification of market niches**, **MSP for coexistence of the fishing, aquaculture and other activities**, and consolidation of the competitiveness of the **Cape Verde fish processing sector**, which is one of the pillars of its economy. Projects should be conducted to address those needs. In relation to capacity building, the sea basin is progressing in terms of participating in the EU institutional structures that may allow a better development of the fisheries activity. A remarkable case is that of participation of the fishing sectors of Madeira, Azores and Canary Islands in the SWW AC. The OR sectors are also engaged in the implementation of the ORs' AC. Improvement of **education and training** is required in the entire sea basin and is especially relevant for Cape Verde. This constitutes an important issue to be dealt with through cooperation, within the MAC program and also at intergovernmental level. Access to finance for investments in the framework of the structural funds for fisheries is available only for the ORs. **Data collection** faces many limitations in the sea basin and especially in relation to **socioeconomic variables**. **Data for scientific purposes** should be reinforced in the case of Cape Verde. Besides cooperation in the framework of the MAC program the private and public sectors are active in initiatives

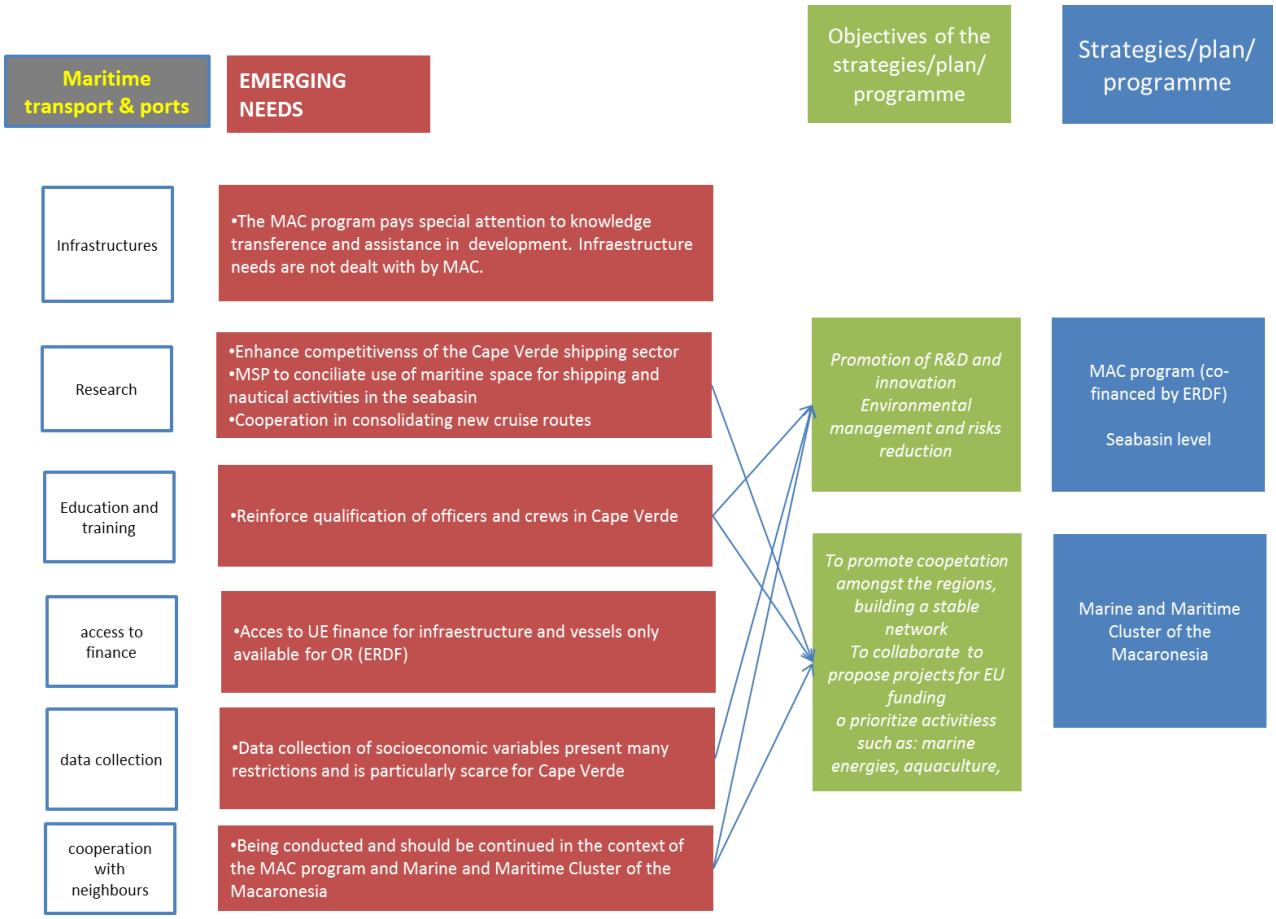
related to the Marine and Maritime Cluster of the Macaronesia (MMCM), where cooperation in the field of aquaculture has particular interest.

Figure 2 — Logical diagram for desalination of seawater



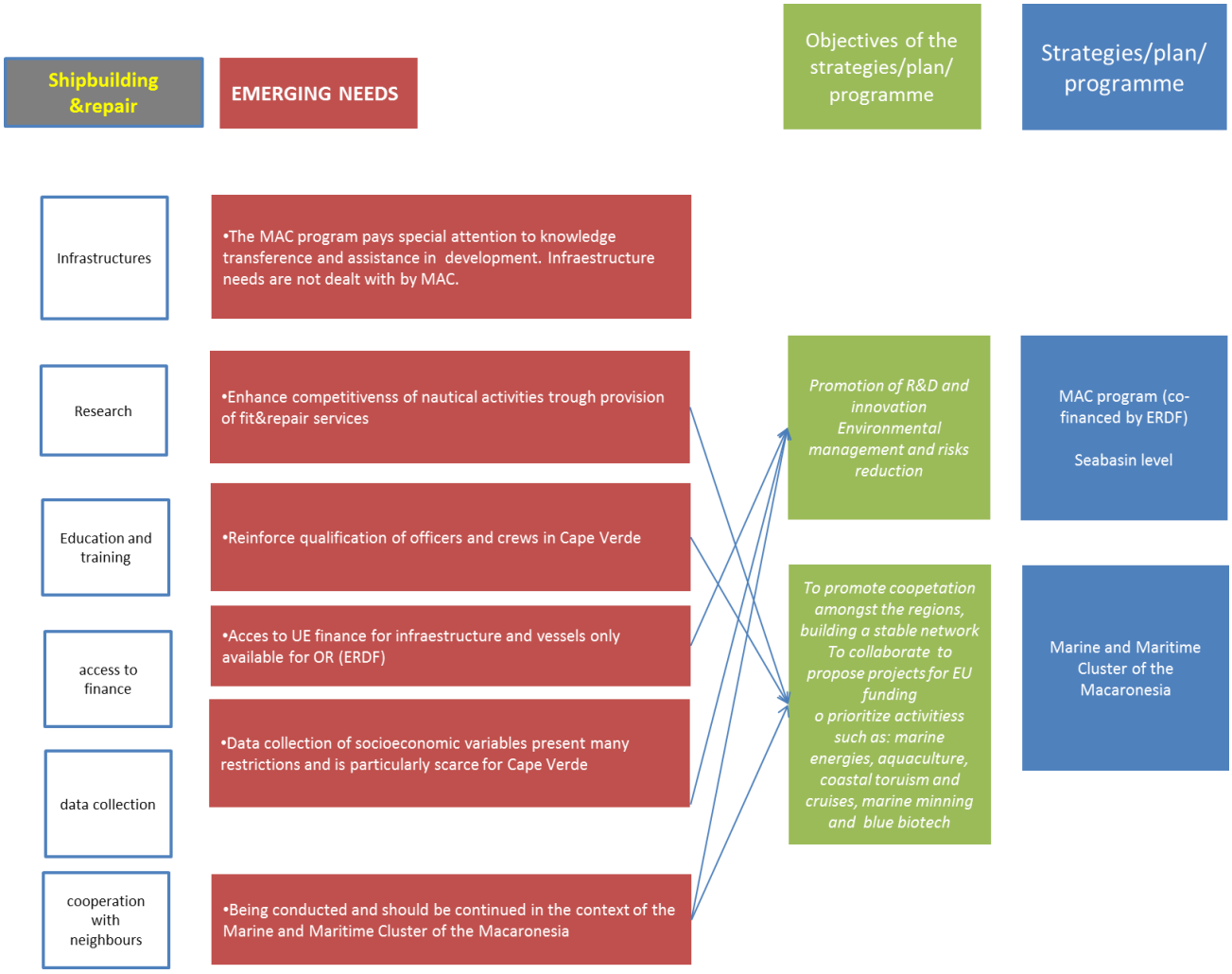
This is an activity where cooperation is basically conducted between Canary Islands and Cape Verde. **Infrastructures are not funded in the framework of cooperation mechanism** such the program MAC. Cape Verde may find this funding in the realm of third countries' aid/finance or through the EDF. R&D and innovation needs are those highlighted in the Canary Islands' report i.e. improvement of the water quality and **use of renewable energies in the production process** are specially required. Transference of the outcomes of research to Cape Verde is of key relevance. In relation to capacity building the assistance of the Canary Islands in desalination project includes advice in the planning of the activity in order to providing an organizational basis for the development of the activity. Regarding employment and training, the Canary Islands have the capacities to provide training to the Cape Verde's personnel. Access to finance to Cape Verde is not provided by the ERDF but by third countries and EDF. **Data collection** faces many limitations in the sea basin and especially in relation to **socioeconomic variables**. Cooperation is conducted in the framework of the MAC program and also being addressed at the level intergovernmental agreements.

Figure 3 — Logical diagram for maritime transport and ports



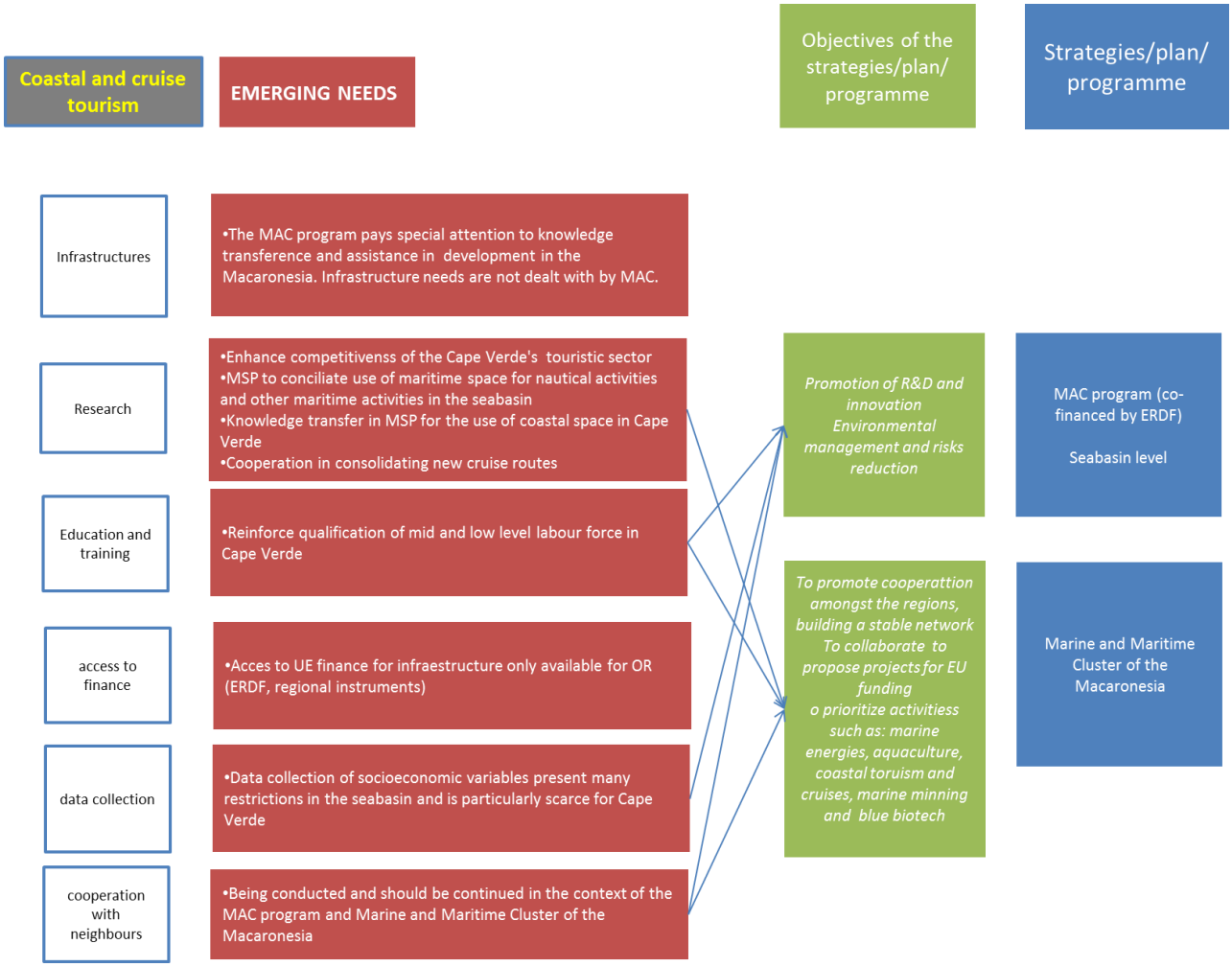
In the present study, infrastructures needs are discussed at the level of the OR analysis. At the level of Macaronesia sea basin, cooperation in knowledge transference and assistance for development is co-funded by the MAC program. As for R&D and innovation, developing MSP for activities such as maritime transport, nautical activities and cruises arises as a key need that should be addressed under the umbrella of the MAC program. Regarding capacity building institutional frameworks are in place at the level of the ORs. At the level of the sea basin, the marine and maritime sectors are organizing themselves in the MMCM. This allows preparing joint proposals and interacting with authorities. Education and training is required especially in the case of **Cape Verde**, where there is an **apparent lack of educational and training offer** to respond to the needs of the shipping sector. Cooperation could be conducted and is, in fact, being developed in the framework of MAC program and initiatives led by the main ports of the Canary Islands. Access to finance for investments is not available for the entire sea basin. The ERDF for infrastructure and vessels apply only for the ORs. Data collection faces many limitations in the sea basin and, especially, in relation to socioeconomic variables. Besides the usual cooperation in the framework of the MAC program the private and academic sectors are active in initiatives related to the MMCM, where cooperation in the field of maritime transport has particular interest.

Figure 4 — Logical diagram for shipbuilding and repair



In this study infrastructures needs are discussed at the level of the OR analysis only. At the level of sea basin, cooperation in knowledge transference and assistance for development is co-funded by the MAC program. As for R&D and innovation, **improvements in fit and repair of nautical equipment** in the EU regions could substantially contribute to the development of the activity in the entire sea basin and, in particular, in Cape Verde where knowledge transference and infrastructures are needed. Regarding capacity building institutional frameworks are in place at the level of the ORs. At the level of the sea basin, however, the marine and maritime sectors are organizing themselves in the MMCM. This allows preparing joint proposals and interacting with authorities. Education and training is required especially in Cape Verde, where there is a little training offer to respond to the needs of the nautical sector requirements. It is worth mentioning that the transit of people between Cape Verde and the EU ORs requires a visa. This fact may limit mobilization of people between these regions. This has implications for tourism and other activities such as training of Cape Verde workers. Access to finance for investments is not available for the entire sea basin. The ERDF and other regional instruments apply only for the EU's ORs. Data collection faces many restrictions in the sea basin and, especially, in relation to socioeconomic variables. Besides the usual cooperation in the framework of the MAC program the private and public sectors are active in initiatives related to the MMCM, where cooperation in the field of ship repair and nautical activities has particular relevance.

Figure 5 — Logical diagram for coastal and cruise tourism

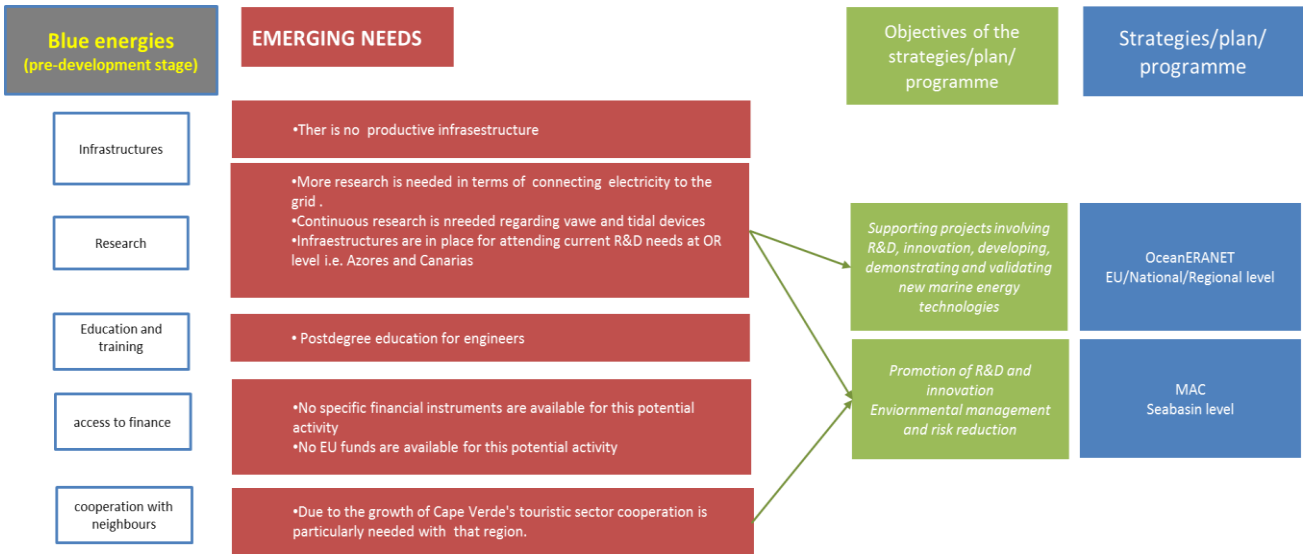


In the present study, infrastructures needs are dealt with at the level of the OR analysis. At the level of Macaronesia sea basin cooperation in knowledge transference and assistance for development is co-funded by the MAC program. As for R&D and innovation, developing MSP for nautical activities and other maritime activities is as a key need that should be addressed under the umbrella of the MAC program or other sources of funding. Regarding capacity building institutional frameworks are in place at the level of the ORs. At the level of Macaronesia, however, the marine and maritime sectors are organising themselves in the MMCM. In addition, the **regions are cooperating in the development of the cruise activity** through the Association cruises in the Atlantic Islands. Education and training is required especially in the case of Cape Verde, where there is an apparent lack of **educational and training offer to respond to the needs of the tourism** sector at all levels of qualification. Cooperation could be conducted and is in fact being developed in the framework of MAC and other initiatives led by the OR governments and the MMCM. Access to finance for investments is not available for the entire sea basin. The ERDF for infrastructures and regional instruments are available only for the ORs. **Data collection** faces many restrictions in the sea basin and, especially, in relation to **socioeconomic variables**. Efforts should be conducted to create a socioeconomic observatory for the blue economy for the sea basin, comprising the marine and maritime activities discussed in this report. Besides cooperation in the framework of the MAC program the private and public sectors are active in initiatives related to the MMCM and, in the case of nautical activities, in the framework of the project Technological Cluster of the Macaronesia, which is led by the academic sector.

Figure 6 — Logical diagram for blue biotechnology (pre-development stage)

This activity is at pre-development stage. R&D efforts are being conducted at the level of the ORs, especially in Madeira and the Canary Islands. The future activity relies on biotechnological developments based on macroalgae and microalgae production. In fact, production of macroalgae and microalgae are taking place in Madeira and the Canary Islands, respectively, as part of the aquaculture sectors. There is a potential to add value to the microalgae production trough extraction of bioactive compounds to respond to a growing demand for a large variety of industrial uses. This faces the constraint of large investments in equipment. Strengthening R&D, research infrastructures and training are envisaged in the program Marine Biotech (ERA NET). This initiative has the potential to become a key driver in for EU ORs. Access to finance is constrained being private investments neither supported by specific financial instruments nor by public funding. Regarding education, efforts need to be conducted at regional level for educating specialised researchers, engineers and technicians. In relation to cooperation at the sea basin level the MAC program is a key driver to promote R&D and training of researchers from the diverse regions in the Macaronesia basin.

Figure 7 — Logical diagram for blue energies (pre-development stage)



The blue energy sector is at pre-development stage in the sea basin. In Azores, there is one wave energy plant. This central was the first in the world to be connected to an electricity distribution network. Its functioning is not continuous. Numerous technical and structural problems require a constant and important financing. Although sound R&D is being conducted in the Canary Islands the production of electricity at commercial level may take some few years more to start. Thus, infrastructures to produce electricity at commercial level are not yet in place in the sea basin. Wave and offshore wind energies are likely the modalities of blue energy with highest potential. R&D in these fields is being conducted by PLOCAN and ITC in the Canary Islands. In the future, this activity is likely to provide jobs for highly qualified professionals at both university and technical level. Postgraduate studies for engineers are likely to be needed. The OceanERANET program addressed R&D needs and aims at demonstrating the validity of technological developments by funding research initiatives. The Canary Islands R&D system is part of this consortium. There are neither private financial instruments nor public funds for the development of the commercial production of marine energy. Only R&D is being funded by EU funds and national and regional funds. In the framework of the program MAC, cooperation with neighbors is being carried out regarding renewable energies in general. Cooperation with Cape Verde is particular relevant due to the increasing needs of energy in that region due to the growth of tourism and needs of water and energy supply.

13 Position of the EU ORs in the Macaronesia sea basin

A comparative analysis of the relation between the ORs and Cape Verde shows that, in general, the EU regions have many assets making them competitive in relation to the African country but also representing opportunities for cooperation at the entire sea basin level. Amongst these assets:

- knowledge, expertise and skills for maritime activities (workforce and educational programs at high, medium and low technical level);
- infrastructures (ports, accommodations, R&D infrastructures, amongst others);
- public services e.g. health, security and transport;
- a strong institutional framework, connected to international fora trough the national and EU networks;
- EU funds for economic development and infrastructures at the OR level;
- a private sector conscious of the opportunities offered by blue growth for the development of the ORs.

It is worth mentioning that Cape Verde is the only republic in the sea basin and, as such, is independent in making decisions in strategic matters such as education, monetary and fiscal policies, management of natural resources, international agreements, amongst others. In contrast, the ORs depend on national and EU decisions for many issues. The fact that Cape Verde is a republic may facilitate coordination in cooperation in the sea basin. A SWOT analysis of the position of the ORs in the Macaronesia sea basin is provided below. In general, it seems that the regions in the sea basin may obtain more advantages of cooperating rather than of competing.

Table 6 — SWOT analysis of the position of the ORs in the Macaronesia sea basin

Strengths	Weaknesses
Blue economy competitiveness	
Public and private strategies specially developed to promote the growth of the blue economy in the OR with a view of strengthening cooperation with Cape Verde. The ORs private sector leads initiatives in the sea basin i.e. Maritime Cluster of the	Double insularity raises transport costs that impede developed of commercial exchange from all islands in the ORs with Cape Verde. Competition with Cape Verde in the sea basin, especially regarding coastal tourism. High dependency on fossil energies is a

Strengths	Weaknesses
Macaronesia. The ORs academic sector leads initiatives in the sea basin i.e. project Technological Platform of the Macaronesia. A competitive R&D sector with an advanced university and research system. The ORs port infrastructures and expertise are able to provide support to the growth of the shipping and cruise activity in the entire sea basin. Port infrastructures and logistics are in place to provide humanitarian support to African nations.	common problem for the entire sea basin. Water scarcity requires expansion of the desalination activity in the Canary Islands and Cape Verde, which is currently strongly linked with use of fossil energies.
Labour market in the maritime sector	
The ORs have a competitive advantage in terms of human capital, training and education infrastructures (e.g. Azores) to respond up to the new labour needs in the maritime activity, e.g. shipping. The ORs seems to have a large potential labour market for maritime activities, especially for shipping, ship repair and coastal and maritime tourism. Cooperation between the ORs and Cape Verde in nautical activities and cruise tourism is being carried out.	The sea basin has a lack of training offer for low and mid-level technical workers to quickly respond to emerging needs. Seasonality of the touristic activity impedes consolidation careers in the sector. This is a weakness in the entire sea basin. Lack of language skills may harden communication and interaction amongst the regions in the sea basin. Weak generational relief in shipping and fishing (on board activities). This is reported specially in relation to the Canary Islands, where generational relief seems more acute than in the other archipelagos.
Maritime research and innovation	
The ORs enjoy a highly advanced R&D system with a vocation for leadership in the sea basin and even beyond. Public finance mechanisms in the ORs are in place to support private initiatives in innovative projects and private R&D. Infrastructures are in place to provide support to R&D and innovation initiatives in the sea basin and beyond (e.g. Canary Islands). The archipelagos are strategically located to turn into a natural lab for R&D, having the human capacity and the infrastructures for a diversity of blue activities, such as, for example, blue energies.	High dependency on EU funding instruments, which cannot provide continuity to R&D projects in the OR. Many OR's researchers and, in general, graduate and post-graduates have migrated to developed countries due to lack of professional opportunities and low salaries. Thus, there is an exit of talent that could otherwise be employed to trigger development in the entire sea basin.

Opportunities	Threats
Blue economy competitiveness	

Opportunities	Threats
Increasing demand of touristic services at the level of the entire sea basin may provide opportunities for the OR in terms of exporting expertise in the touristic sector, while strengthening cooperation in cruise sector. An increasing demand for specialised workers, especially in the nautical activities, including fit and repair of nautical craft, shipping and cruise sectors. These sectors are of key importance for economic development in the sea basin and the ORs are well placed to develop the skills to respond to these emerging needs Increasing opportunities to carry out technological transference to the other regions in particular to Cape Verde e.g. desalination of seawater.	Fluctuations of oil prices may have severe impacts on activities where the ORs are competitive in relation to Cape Verde such as services to offshore activities.
Labour market in the maritime sector	
Increasing demand for activities linked to the nautical sectors in the sea basin. The ORs can contribute by providing a training offer to the other regions due to availability of infrastructures and expertise.	The inability to quickly respond to new labour opportunities (e.g. providing a training and education offer) impedes matching employment supply and offer. This gap is filled with personnel from outside the sea basin.
Maritime research and innovation	
Emerging activities such as marine energies are strongly required to diminish dependency on fossil energies. This provides opportunities for R&D and innovation that can be exported to Cape Verde.	R&D is strongly dependent on EU aid and linked to the life span of EU funded projects. This affects R&D in the entire sea basin.

14 Conclusions on concrete measures by type and timeframe

By analysing the Logical diagrams above, and considering all findings reported in the previous sections of the report, the following 'legislative', 'non-legislative' and 'financial' conclusions emerged. For each of them, the following details are provided:

- possible time frame for fulfilling the potential (short/medium term or long term);
- stakeholders that should activate the measure;
- possible best practices experimented in other territories (i.e. cross-fertilisation possibilities);
- measures for achieving/implementing them.

- ***Fisheries and aquaculture***

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
LEGISLATIVE	IUU fishing is a problem for the Macaronesia region. Cooperation in MCS at sea basin level seems necessary in coordination with the relevant Regional Fisheries Management Organisations in order to set up such a system in the EEZs, or to coordinate market mechanisms to restrict the entry of IUU products into formal fish markets.	Medium to long term	Regional authorities National authorities Private stakeholders	Port state measures and market restrictions are in place in diverse regions of the world. Efforts are being made especially by the tunas RFMOs. Technological improvements such as aerial vigilance are being used in coastal areas of the Canary Islands.	Joint cooperation between the authorities, fishing sector and navy should be strengthened.
	The emergence of recreational fishing is also becoming a problem for the professional sector. Controlling these activities is particularly hard due to the large number of recreational fisheries and atomization of ports. Besides pressure over the resources, unreported catches may undermine stock assessments. Control of the recreational activities in order to avoid the entry of fish into commercialization channels is required.	Medium to long term	Regional authorities National authorities Private stakeholders	Measures implemented in the Canary Island to control distribution channels have managed to diminish this problem. The New Zealand paper trail allows effectively controlling the fish distribution channels and provides an example of successful implementation of a traceability measure.	A sound study characterising the recreational activity, identifying the main challenges of controlling the activities at sea and access to market of these fishers should be conducted in order to design effective measures to restrict access of these fish into the commercialization channels.
	Coexistence of fishing with other uses of the seas such as shipping and ports, coastal tourism, nautical activities, aquaculture and conservation (MPAs) require a sound Maritime Spatial Planning.	Medium to long term	Regional authorities National authorities Private stakeholders	Not available	A study mapping uses and users of the marine and maritime space and proposing measures to activate maritime governance in the sea basin.

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
	Administrative procedures to obtain aquaculture concessions in the ORs are regarded as too long and may discourage investment. There is a need to optimize administrative processes in order to facilitate investment.	Medium to long term	Regional authorities National authorities Private stakeholders	Not available	A study identifying bottle necks and conflicts of competences in the granting of concessions at OR level.
NON-LEGISLATIVE	The fishing sector faces the problems of remoteness and insularity, which increase transport costs and hinders competitiveness. Promotion of local fish products at sea basin/regional level would reduce dependency on imports, cut down transport costs, and improve food security and value addition.	Medium to long term	Regional authorities National authorities Private stakeholders	Not available	Identification of new market niches at regional and international level, assessing acceptability of new species and products.
	Research on new species, e.g. deep water, as an alternative to reduce pressure over traditional species and to diversify markets and products is required. This is particularly relevant for Cape Verde, where scientific capacity seems limited.	Medium to long term	Regional authorities National authorities Research and academia	Not available	ORs to continue with EU funded projects for cooperation in building of scientific capacity in Cape Verde.
	Broadly speaking the fishing sector faces low generational relief, and lack of professionalization of the fishermen. Low qualification is especially notable In Cape Verde, where training in aquaculture is limited due to lack of specialized schools. These issues hinder the competitiveness of the sectors.	Medium term	Regional authorities National authorities Research and academia	Not available	Promotional campaign to inform youngsters about the opportunities provided by the fishing and aquaculture activities. Training for fishermen in managerial skills and other abilities in order to promote entrepreneurship. Cooperation with Cape Verde in training of personnel and assistance in evaluation the technical viability of a training school.

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
	Pesca-tourism and other diversification for boats activities for crews and boats, e.g. observation of sea mammals, have the potential to create jobs and reduce pressure over the resources. Training for fishermen is required for these activities.	Short term action	Regional authorities National authorities Research and academia	Not available	Training of fishermen and other interested parties in abilities required to conduct these forms of tourism. Cooperation with Cape Verde could be of great benefit for that region. Actions to promote synergies between the fishing and the tourist sector could be promoted in order to identify opportunities.
FINANCIAL	OR fishing and aquaculture sectors are heavily dependent on structural funds. Ad hoc private financial instruments are not provided. Measures at the level of ORs should be carried out to counteract this issue.	Medium term	Regional authorities National authorities Private stakeholders	Not available	This measure is only relevant for the EU's ORs. Governments could assess the possibility to back up investments negotiating preferential interests and guarantees with financial intermediaries.

- ***Desalination***

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
LEGISLATIVE	Collaboration is being strengthened between the Canary Islands and Cape Verde on this matter. Other initiatives are being made in the framework of the EU funded programs. These collaborations should be continued and also involve Madeira.	Short to medium term	Regional authorities National authorities Research and academia	Not available	In progress. Proposals are being made in the framework of the MAC program. Bilateral cooperation between the governments of the Canary Islands and Cape Verde are being conducted.
NON-LEGISLATIVE	Powering of desalination plants with wind energy is an alternative for fossil energy dependency. R&D in this matter should be continued in the Canary Islands and experience transferred to Cape Verde.	Short to medium term	Regional authorities National authorities Research and academia	Not available	R&D to be continued by ITC. Transference of technology to Cape Verde to be made in the framework of EU funded projects.

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
FINANCIAL	No financial conclusions identified				

- ***Maritime transport and ports***

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
LEGISLATIVE	Enhanced collaboration is needed between the regions regarding strengthening logistics, service standards and technological advancements. Cooperation is required between ports administration within the sea basin.	Medium to long term	Regional authorities National authorities Private stakeholders	Not available	To establish a permanent network for collaboration, sharing of knowledge and experience, implementation of common projects, and training of personnel. Ports in the sea basin are already active in cooperation through the MAC program, but there is still need for cooperation.
NON-LEGISLATIVE	The transport and ports activities require qualified personnel, especially in shipping and management logistics. It is reported that the workforce is not sufficiently trained in Madeira. In the case of the Canary Islands it seems that the workforce offer can currently satisfy the needs although it is envisaged that the expansion of activities, e.g. offshore activities, will require more qualified personnel. In Cape Verde the activity is very limited. Efforts should be conducted to promote sea careers.	Medium term	Regional authorities National authorities Private stakeholders	Not available	Development of cooperation to provide a training offer in shipping and logistics management. Cooperation between the ports of the sea basin, including those of Cape Verde will be required to establish a network of ports in the sea-basin. The Maritime Cluster of the Macaronesia is well places to promote these activities.
	A general problem is the lack of generational replacement in sea careers, especially of crews. Actions need to be conducted to counteract this issue.	Medium term	Regional authorities National authorities Private stakeholders	Not available	Develop an information and promotional campaign in order to inform about the opportunities provided by the sea careers.

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
	Large investments have been made to turn OR ports into competitive infrastructures able to take advantage of the strategic location of the archipelagos. However, in the Portuguese ORs improvements on support services linked to ships, freight and crews are needed. In the Canary Islands improvements are needed for facing the increasing demand of offshore services e.g. bunkering. Infrastructures for nautical activities need to be improved to satisfy the growing demand e.g. number and size of moorings.	Medium to long term	Regional authorities National authorities Private stakeholders	Not available	Conduct cooperation in the framework of transnational agreements and maritime clusters activities.
FINANCIAL	No financial conclusions identified.				

- ***Shipbuilding and repair***

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
LEGISLATIVE	No legislative conclusions identified.				
NON-LEGISLATIVE	Due to the emergence of the nautical activities the growth of fit and repair businesses will require port's space reorganisation, enlargement of naval infrastructures and improvements in equipment and material provision.	Medium to long term	Regional authorities Private stakeholders	Not available	This is only relevant at the level of the ORs.

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
	Fit & repair of recreational boats presents good perspectives in the entire sea basin but require continued efforts for improvement of technical skills e.g. synthetic materials.	Medium term	Regional authorities Private stakeholders	At OR level, initiatives have been conducted by the city government of Las Palmas (fibre glass training).	Cooperation could be conducted on technical mid/low level training in repair and maintenance of recreational boats and other marine recreational craft, especially in the use of synthetic materials.
FINANCIAL	No specific instruments are provided for the development of private initiatives. Financial mechanisms should be identified.	Medium term	Regional government Private stakeholders	Not available	Evaluation of EIB financial opportunities in the framework of the LeaderSHIP program.

- ***Coastal and cruise tourism***

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
LEGISLATIVE	New tourism activities, which encompass a more comprehensive utilisation of the marine space and appreciation of its natural resources, e.g. whale and birds watching, diving, exploration of seamounts and reserves, etc., may require agreements at sea basin level.	Medium to long term	Regional authorities	Not available	Coordination at the level of the regions concerned in order to establish knowledge exchange and setting up of good practices for this kind of tourism.
NON-LEGISLATIVE	Qualification is required to satisfy the new needs of the tourism activity. This is required for mid- and low-level personnel. In the case of emerging touristic activities, new qualifications are required for tourist guides and operators in the field of observation of marine life and environment. The growth of the nautical sector requires marina administrators. Promotion of cultural activities in coastal communities requires qualified personnel.	Medium to long term	Regional authorities Private stakeholders	Not available	In the ORs, new needs will require complementary training and certification for new tourist activities in order to develop competitiveness and deter the emergence of informal operators. These initiatives should be transferred to Cape Verde in the framework of the MAC program.

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
	The growth of nautical activities requires port improvements, building of marinas and reorganization of the port space.	Medium to long term	Regional authorities Private stakeholders	Not available	This is only relevant at OR level.
	Improvement of city infrastructures, services, personnel with language abilities and touristic packages are required for cruisers doing a stop in the OR's islands.	Medium to long term	Regional authorities Private stakeholders	Not available	This is only relevant at OR level.
	Water and energy are scarce resources, especially in the Canary Islands and Cape Verde. Continued R&D in blue energies is required to satisfy energy needs and to improve energy efficiency. Cooperation in desalination between the Canary Islands and Cape Verde should be continued.	Long term	Regional authorities Private stakeholders	Not available	These R&D initiatives should be transferred to Cape Verde in the framework of the MAC program.
	Seasonality in coastal and cruise tourism needs to be counteracted by innovative touristic products and new routes. Cooperation at the level of the sea basin needs to be continued. Stretching bonds with touristic operators is required to offer complementary services to cruise stopovers such as excursions and resorts.	Short to medium term	Regional authorities Private stakeholders	Not available	Need to ensure adequate berthing, especially in Azores. Cooperation at the level of the sea basin e.g. Association cruises in the Atlantic Islands, and Maritime Cluster should be reinforced. Identification of synergies with touristic operators and identification of a complementary touristic offer for stopovers.
FINANCIAL	No financial conclusions have been identified				

- ***Blue biotechnologies (pre-development stage)***

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
LEGISLATVE	No legislative conclusion identified.				
NON-LEGISLATIVE	Although this activity has a greater potential in the ORs, significant cooperation could tackle challenges related to R&D, education and training and best practices exchange. Thus, continuous cooperation in the Macaronesia region is required.	Medium to long term	Regional authorities Private stakeholders	Not available	Further develop strategies to address challenges regarding the marine biotechnologies (funding, R&D, training, amongst others). Currently, significant work is being conducted by the research and academia sectors in the context of the MAC program.
FINANCIAL	No financial conclusion identified.				

- ***Marine renewable energies (pre-development stage)***

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
LEGISLATIVE	No legislative conclusion identified.				
NON-LEGISLATIVE	Although this activity has a greater potential in the ORs, significant cooperation could tackle challenges related to R&D, education and training and best practices exchange. Thus, continuous cooperation in the Macaronesia region is required.	Medium to long term	Regional authorities Private stakeholders	Not available	Further develop strategies to address challenges regarding the blue energies (funding, R&D, training, amongst others).
FINANCIAL	No financial conclusion identified.				

List of interviewees

Organisation	Maritime activities concerned	Number of interviewees
Direção Regional dos Assuntos do Mar	All maritime activities	1
Direção Regional da Economia e Transportes	Ports, cruising, ferry transport, Nautical sports, air transport	1
Secreteria Regional dos Assuntos Parlamentares e Europeus	All maritime activities	2
Observatório Oceânico da Madeira	Fisheries and aquaculture	1
Direção Regional de Pescas	Fisheries and aquaculture	3
Fundación Puertos de Las Palmas	Ports; international cooperation on port matters	1
Grupo de Acción Costera de Las Palmas	Fishing Local Action Group (FLAG)	1
Instituto Tecnológico de Canarias (ITC)	R&D and innovation on renewable energies, desalination, aquaculture, and others.	2
Ayuntamiento de Las Palmas	Promotion of marine and maritime activities	1
Cluster Marítimo de Canarias	Shipping, ports, aquaculture, shipyards, industrial fishing, nautical activities	1
Centro Tecnológico de Ciencias Marinas (CETECIMA)	R&D and innovation on aquaculture, nautical sector, and others.	2
Gobierno de Canarias - Agencia Canaria de Investigación, Innovación y Sociedad	Government. Agency in charge of the implementation of the RIS3	2
Universidad de las Palmas de Gran Canaria	Education, R&D and innovation, aquaculture, algae research	5
Plataforma Oceanográfica de Canarias (PLOCAN)	Research, development and innovation in marine energies, aquaculture, operational oceanography	1
Federación del Metal de Tenerife (FEMETE)	Private federation of the metal sector, including naval sector; it also provides technical training.	4
Asnáutica	Nautical activities	1
Cabildo de Tenerife - Parque Científico y Tecnológico	Research, development and innovation on technological sectors in general	1
Universidad de La Laguna	Coordination of the tecnological platform of the Macaronesia project	1
Instituto Español de Oceanografía (IEO)	Marine and fisheries research	1
ALGALIMENTO S.L	Algae production	1
Elittoral	Marine consultant	1
Cabildo de Gran Canaria - Sociedad de Promocion Económica de Gran Canaria	Island's goverment. Promotion of innovation and technology	1
Sensorlab SL	Environmental and marine technology	1

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